

(12) **United States Patent**
Evenhaim et al.

(10) **Patent No.:** **US 12,406,754 B1**
(45) **Date of Patent:** **Sep. 2, 2025**

(54) **SYSTEMS AND METHODS FOR ANALYZING LONGITUDINAL HEALTH INFORMATION AND GENERATING A DYNAMICALLY STRUCTURED ELECTRONIC FILE**

(71) Applicant: **Veeva Systems Inc.**, Pleasanton, CA (US)

(72) Inventors: **Asaf Evenhaim**, New York, NY (US);
Lindsey Lofgren, New York, NY (US);
Myrto M. Miltiadous, Astoria, NY (US)

(73) Assignee: **Veeva Systems Inc.**, Pleasanton, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 39 days.

(21) Appl. No.: **18/045,550**

(22) Filed: **Oct. 11, 2022**

Related U.S. Application Data

(63) Continuation-in-part of application No. 17/472,360, filed on Sep. 10, 2021, now Pat. No. 11,551,794.

(51) **Int. Cl.**
G06F 16/00 (2019.01)
G06F 16/9535 (2019.01)
G06F 16/955 (2019.01)
G16H 10/60 (2018.01)

(52) **U.S. Cl.**
CPC **G16H 10/60** (2018.01); **G06F 16/9535** (2019.01); **G06F 16/955** (2019.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

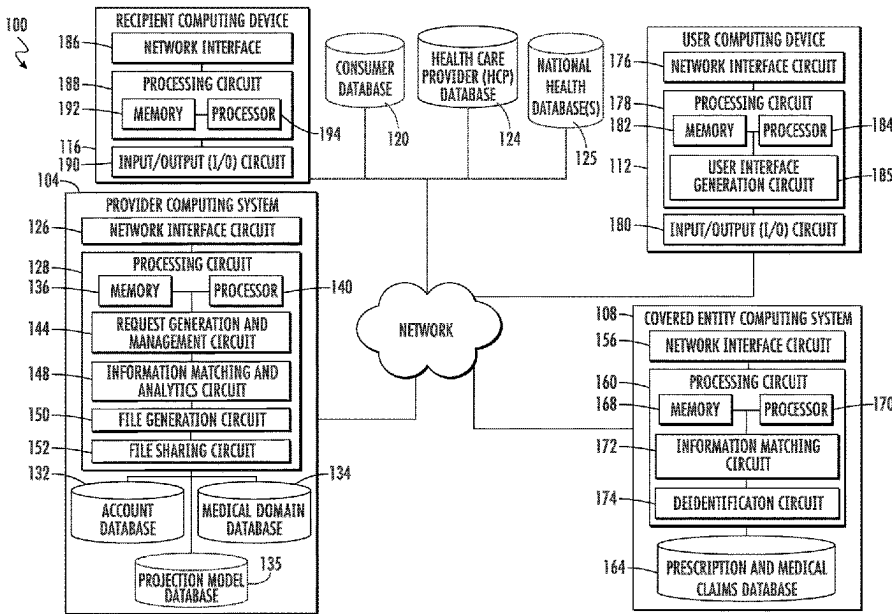
References Cited			
U.S. PATENT DOCUMENTS			
8,595,206	B1	11/2013	Ansari et al.
8,918,385	B1	12/2014	Ansari et al.
10,185,809	B1 *	1/2019	Zhou G16H 40/20
10,231,077	B2	3/2019	Raduchel et al.
10,553,307	B2	2/2020	Connely, IV et al.
10,586,614	B1	3/2020	Cai et al.
10,755,812	B1	8/2020	Cai et al.
10,769,181	B2	9/2020	Bari et al.
10,937,531	B1	3/2021	Wang et al.
11,107,578	B2	8/2021	Nag
11,551,794	B1	1/2023	Lofgren et al.
11,575,673	B2	2/2023	Padmani et al.
11,783,922	B1	10/2023	Tong et al.
11,868,506	B2	1/2024	Johnson et al.
11,948,112	B2	4/2024	Crooks et al.
(Continued)			

Primary Examiner — Tony Mahmoudi
Assistant Examiner — Suman Rajaputra

(57) **ABSTRACT**

A method for generating a dynamically structured electronic file associated with a defined location. The method includes receiving health information from a covered entity computing system and a file generation request from a user computing device. The method further includes combining HCP information with the health information and determining multiple digitally projected values using at least one digital projection model. The method further includes generating the electronic file including at least one digitally projected value and at least a portion of the HCP location information. The electronic file is generated as the file type identified by the file generation request, and is structured based on at least one of the file generation request or the file destination.

19 Claims, 17 Drawing Sheets



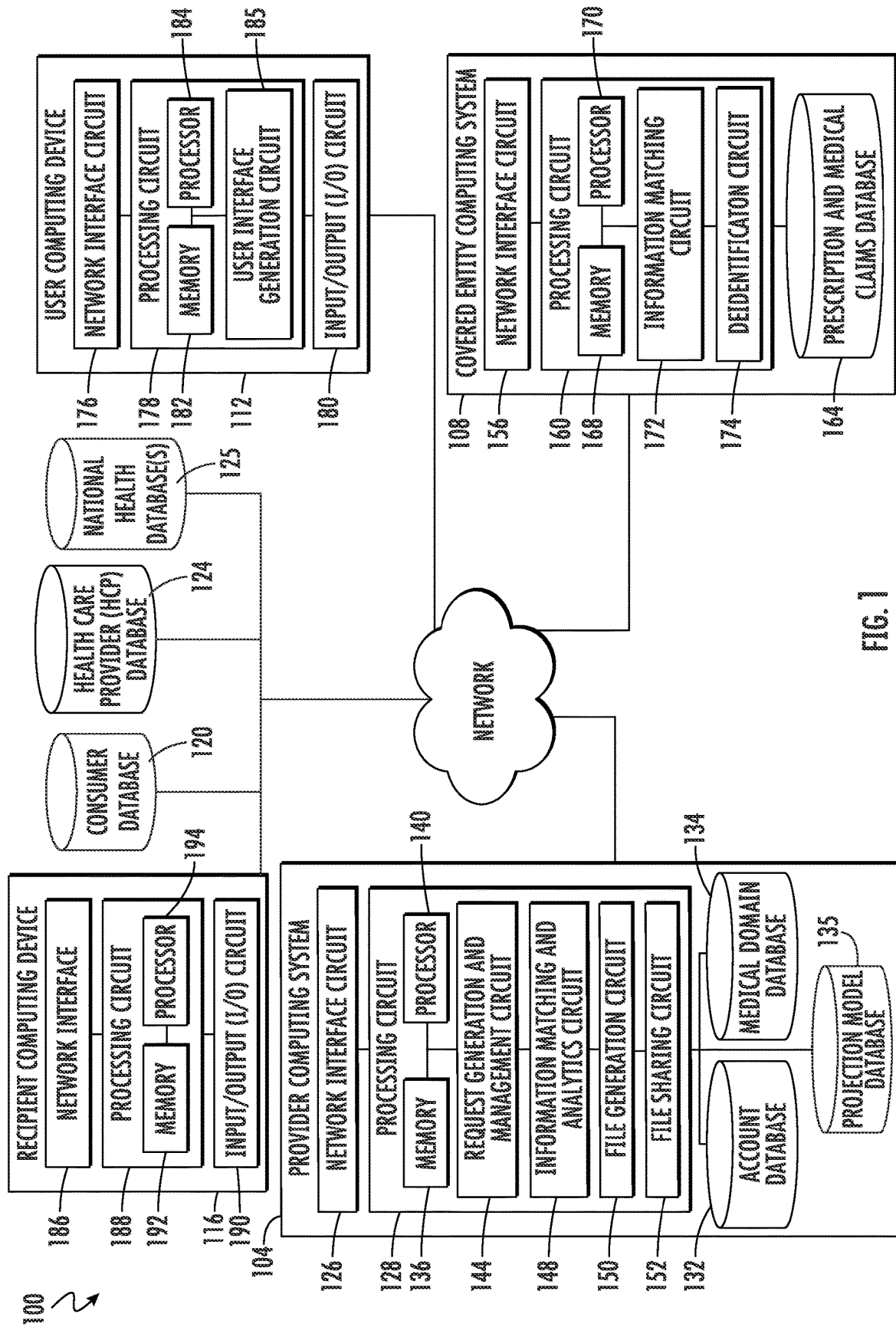
(56)

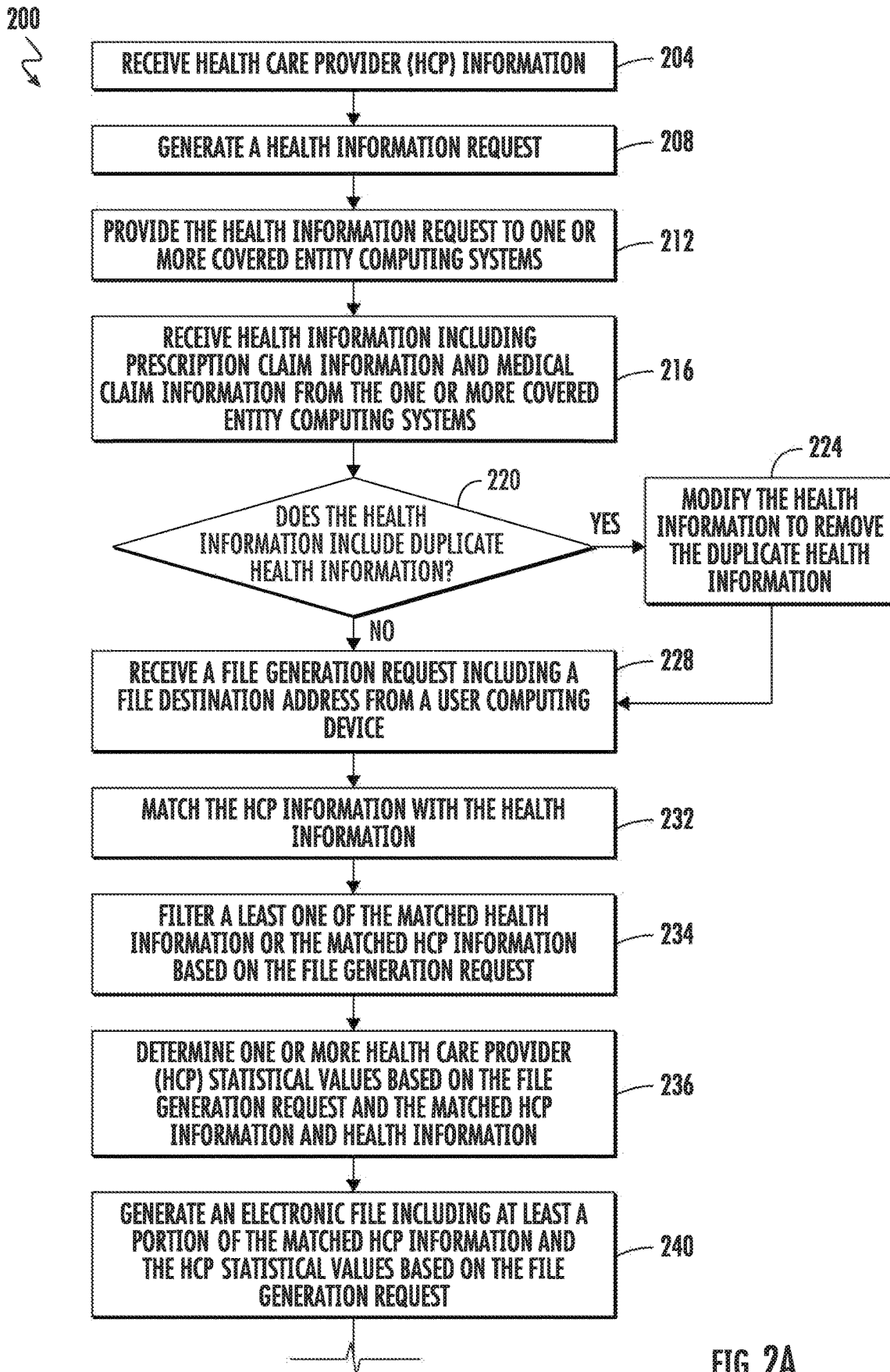
References Cited

U.S. PATENT DOCUMENTS

11,989,326	B2	5/2024	Levy et al.		
12,094,588	B1	9/2024	Hermey et al.		
12,248,526	B1	3/2025	Pushparajah et al.		
2002/0111833	A1	8/2002	Dick		
2002/0116227	A1	8/2002	Dick		
2002/0165736	A1	11/2002	Tolle et al.		
2008/0300902	A1 *	12/2008	Smith	G16H 20/10	
				705/1.1	
2008/0306872	A1	12/2008	Felsher		
2009/0048877	A1	2/2009	Binns et al.		
2010/0131284	A1 *	5/2010	Duffy	G16H 50/20	
				705/7.33	
2010/0241595	A1	9/2010	Felsher		
2010/0324936	A1 *	12/2010	Vishnubhatla	G06Q 40/08	
				715/810	
2011/0112862	A1	5/2011	Yu		
2011/0112970	A1	5/2011	Yu		
2011/0119088	A1	5/2011	Gunn		
2011/0191343	A1	8/2011	Heaton et al.		
2012/0191749	A1	7/2012	New et al.		
2013/0073313	A1	3/2013	Christakis et al.		
2014/0129256	A1 *	5/2014	Veren	G06Q 10/10	
				705/3	
2014/0136237	A1	5/2014	Anderson et al.		
2014/0298030	A1	10/2014	Akiyama et al.		
2015/0046192	A1	2/2015	Raduchel		
2015/0112979	A1	4/2015	Vittorio		
2015/0347599	A1	12/2015	McMains et al.		
2016/0147945	A1	5/2016	MacCarthy et al.		
2016/0210427	A1	7/2016	Mynhier et al.		
2016/0358295	A1	12/2016	Hefley et al.		
2017/0004266	A1	1/2017	Vanderhoef		
2017/0161439	A1 *	6/2017	Raduchel	G16H 10/60	
2018/0121614	A1	5/2018	Connely, IV et al.		
2018/0294048	A1	10/2018	Blumenthal et al.		
2018/0322946	A1	11/2018	Ika et al.		
2019/0252049	A1	8/2019	Fotsch et al.		
2019/0304578	A1	10/2019	Kain et al.		
2020/0005900	A1	1/2020	Cha et al.		
2020/0035341	A1	1/2020	Kain et al.		
2020/0202040	A1	6/2020	Johnson et al.		
2020/0234826	A1	7/2020	Said		
2021/0019296	A1	1/2021	Petersen et al.		
2022/0117692	A1	4/2022	Sellberg et al.		
2022/0270129	A1	8/2022	Dakic et al.		
2022/0270729	A1	8/2022	Saliman et al.		
2022/0319643	A1	10/2022	Skees et al.		
2022/0319697	A1	10/2022	Morris et al.		
2023/0104655	A1	4/2023	Amarasingham et al.		
2023/0110360	A1	4/2023	Sanae et al.		
2023/0144503	A1	5/2023	Powers et al.		
2023/0148326	A1	5/2023	Papel et al.		
2023/0273848	A1	8/2023	Vera-Ciro et al.		
2023/0282322	A1	9/2023	Rajkumar et al.		
2023/0368878	A1	11/2023	Molenda		
2023/0385450	A1	11/2023	Bessette et al.		
2024/0013899	A1	1/2024	Youngblood et al.		
2024/0105289	A1 *	3/2024	Khan	G16H 15/00	
2024/0112771	A1	4/2024	Shelton, IV et al.		
2024/0127932	A1	4/2024	Bongiovanni et al.		
2025/0006352	A1	1/2025	Schoenberg		
2025/0132043	A1	4/2025	Ouimet		
2025/0166762	A1	5/2025	Barkol et al.		
2025/0190621	A1	6/2025	Boominathan et al.		

* cited by examiner





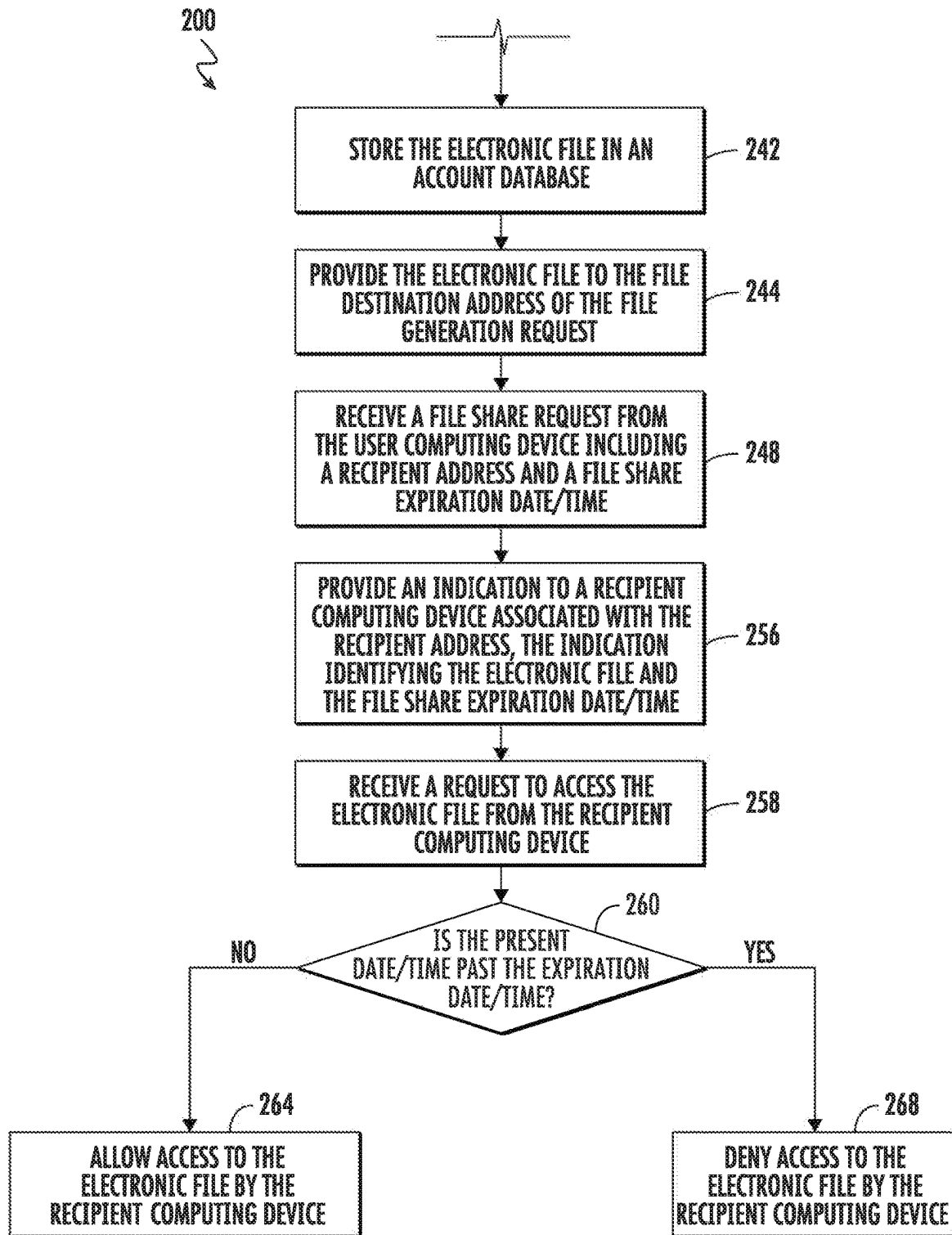


FIG. 2B

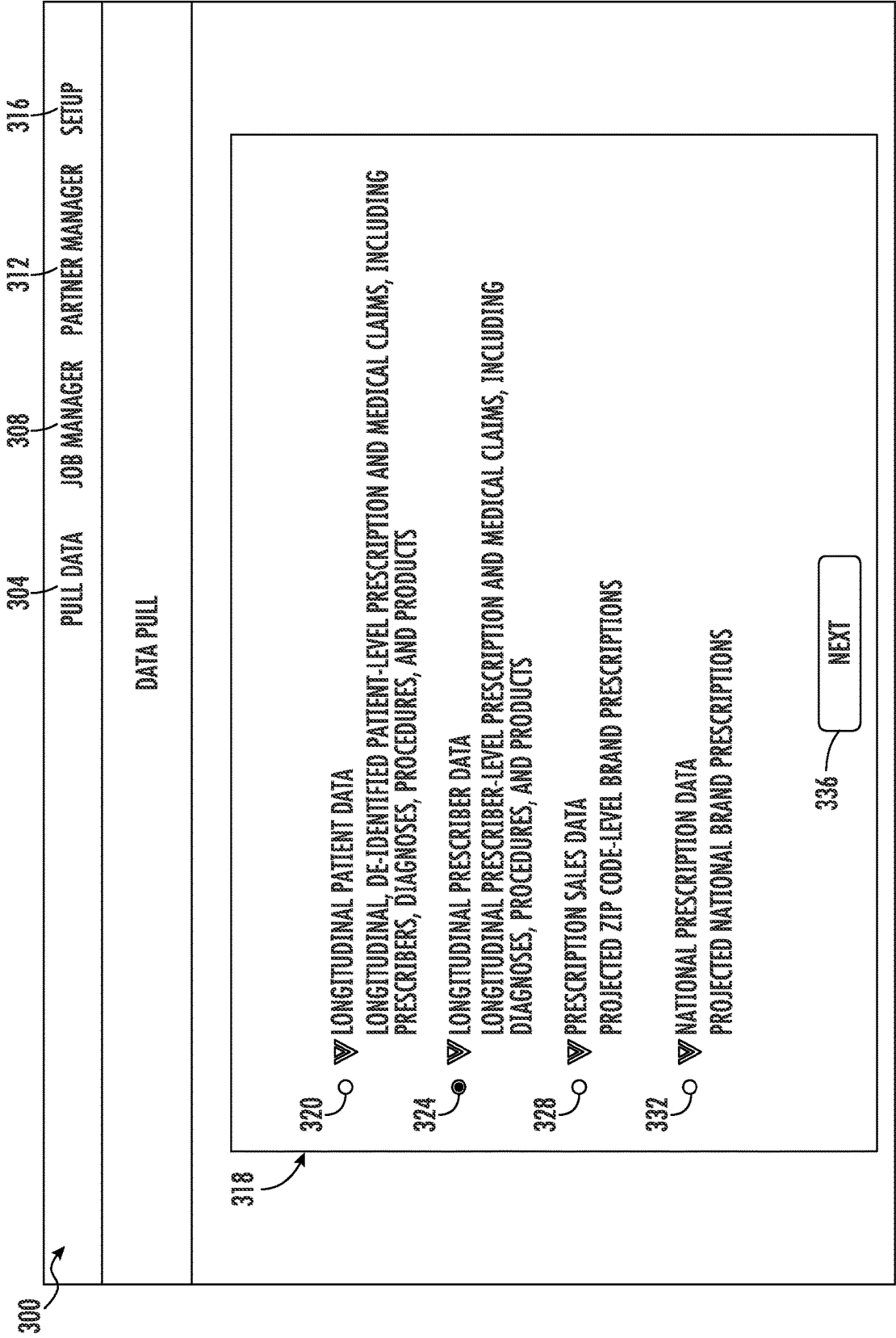


FIG. 3

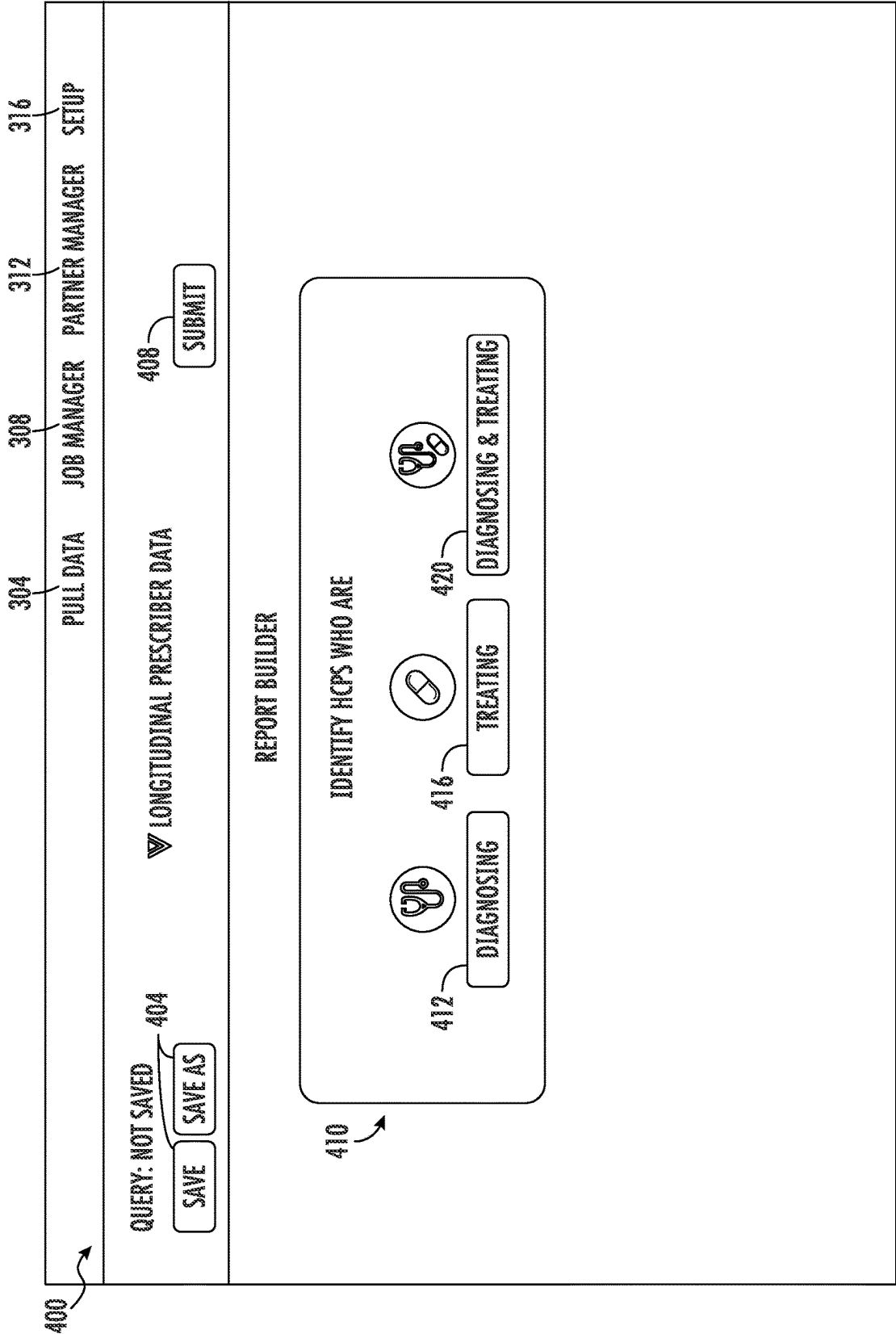


FIG. 4A

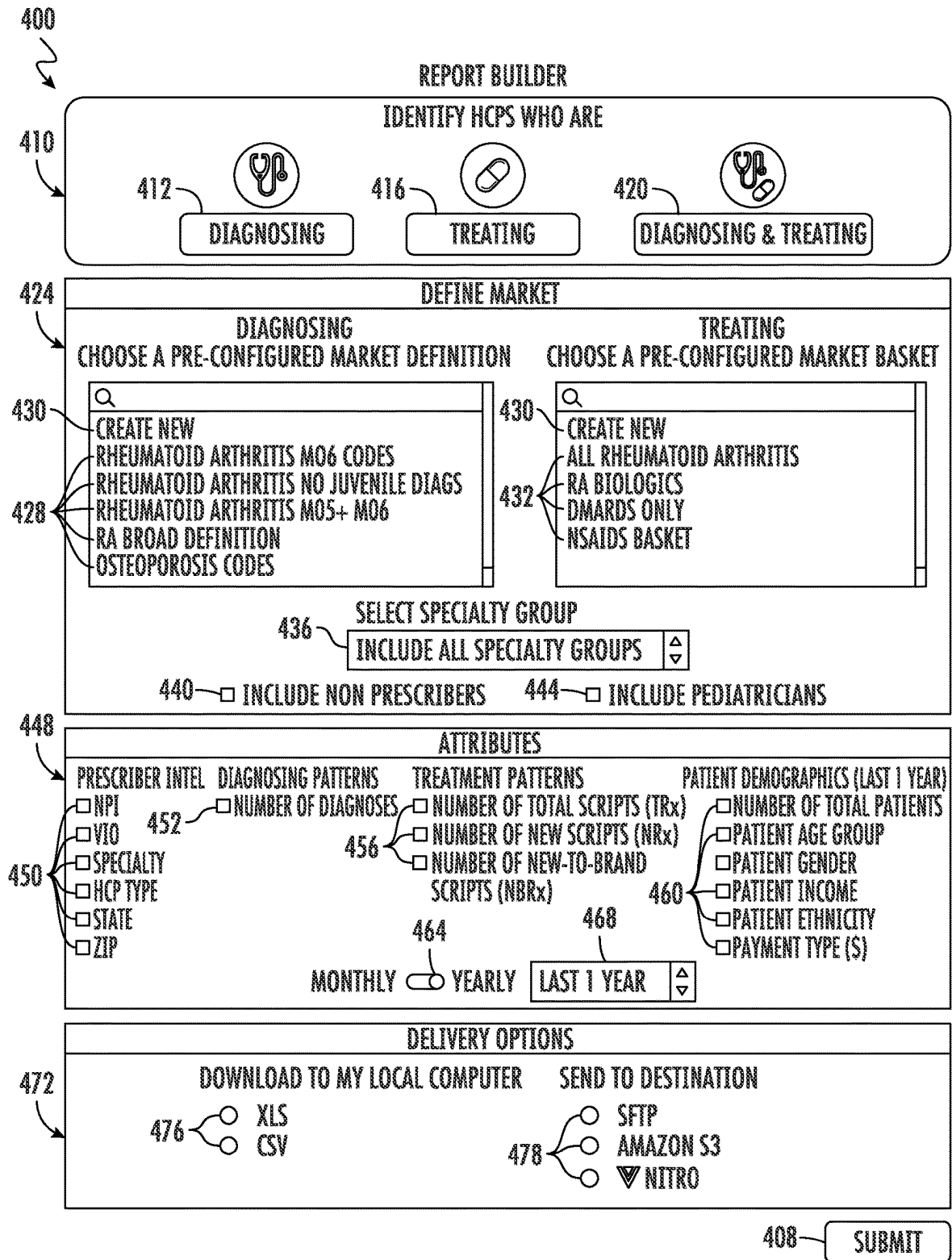


FIG. 4B

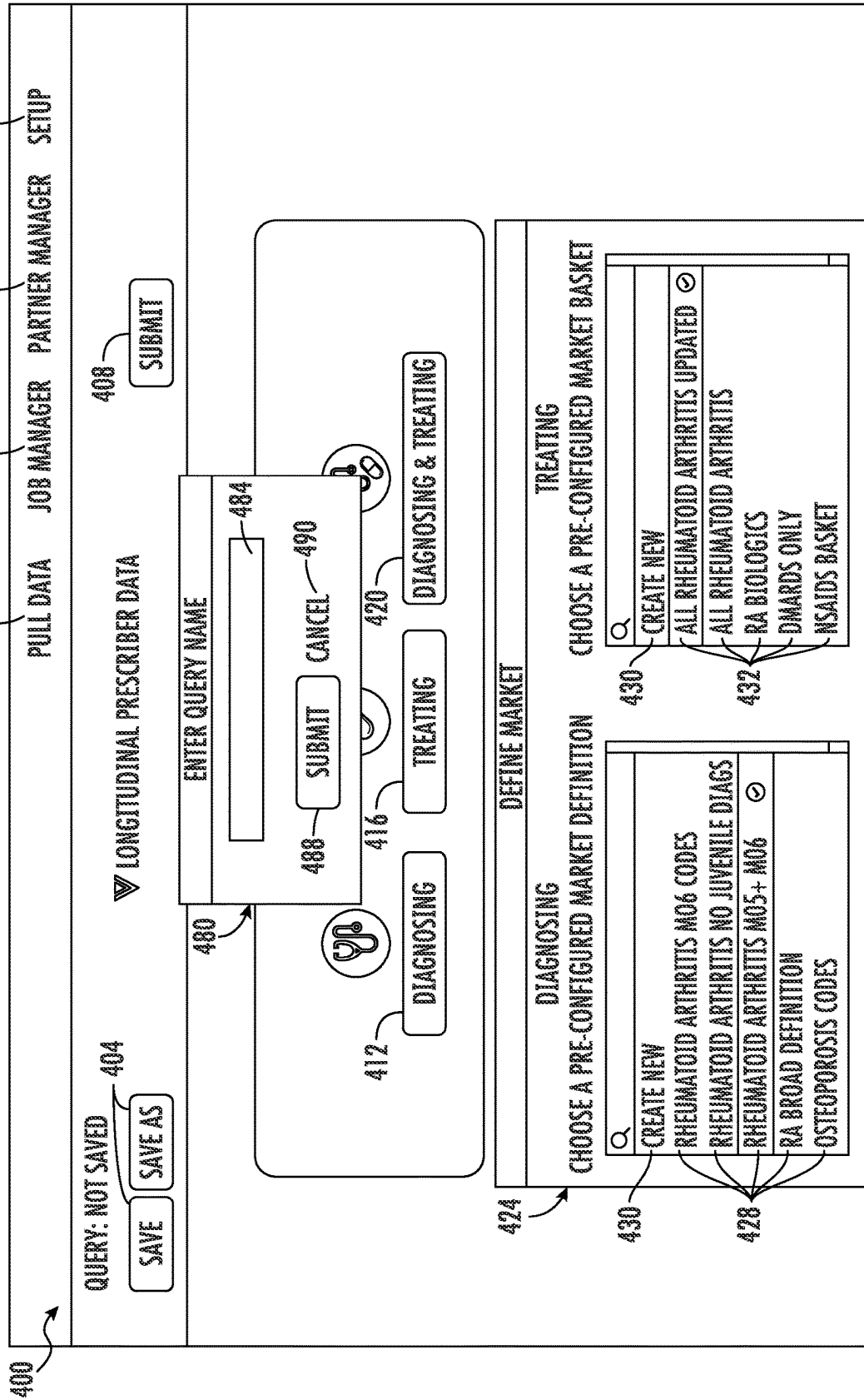


FIG. 4C

LONGITUDINAL PRESCRIBER DATA

QUERY: NOT SAVED SAVE SAVE AS CANCEL DONE

MARKET BASKET ALL RHEUMATOID ARTHRITIS

SAVE DEFINITION AS — 534

DRUG A GO

FILTER BY	DRUG DESCRIPTION	NDC	DRUG COMPOUND/BRANDED/GENERIC	COMPANY NAME	STRENGTH DOSE FORM	ANNUAL ADD ALL (Y/N)
BRAND/Generic	DRUG A	NO GENERIC EQUIVALENTS				
☐ BRAND	DRUG A	12836287920	EUFLEXIMAL	BRAND	VERTO BIOPHARMA 5 mg TABLET	12938
☐ Generic	DRUG A	12836287928	EUFLEXIMAL	BRAND	VERTO BIOPHARMA 10 mg TABLET	8931
Dose Form	DRUG A	31827691189	EUFLEXIMAL	BRAND	VERTO BIOPHARMA 1 mg/mL SOLUTION	10012
☐ TABLET	DRUG A	31827391314	EUFLEXIMAL	BRAND	VERTO BIOPHARMA 5 mg/mL SOLUTION	2839
☐ SOLUTION	DRUG A	31827397598	EUFLEXIMAL	BRAND	VERTO BIOPHARMA 10 mg/mL SOLUTION	<100

ADDED PRODUCTS-7 REMOVE ALL ☒

DRUG B ☒ DRUG C ☒ DRUG D ☒ DRUG E ☒ DRUG F ☒ DRUG G ☒ DRUG H ☒



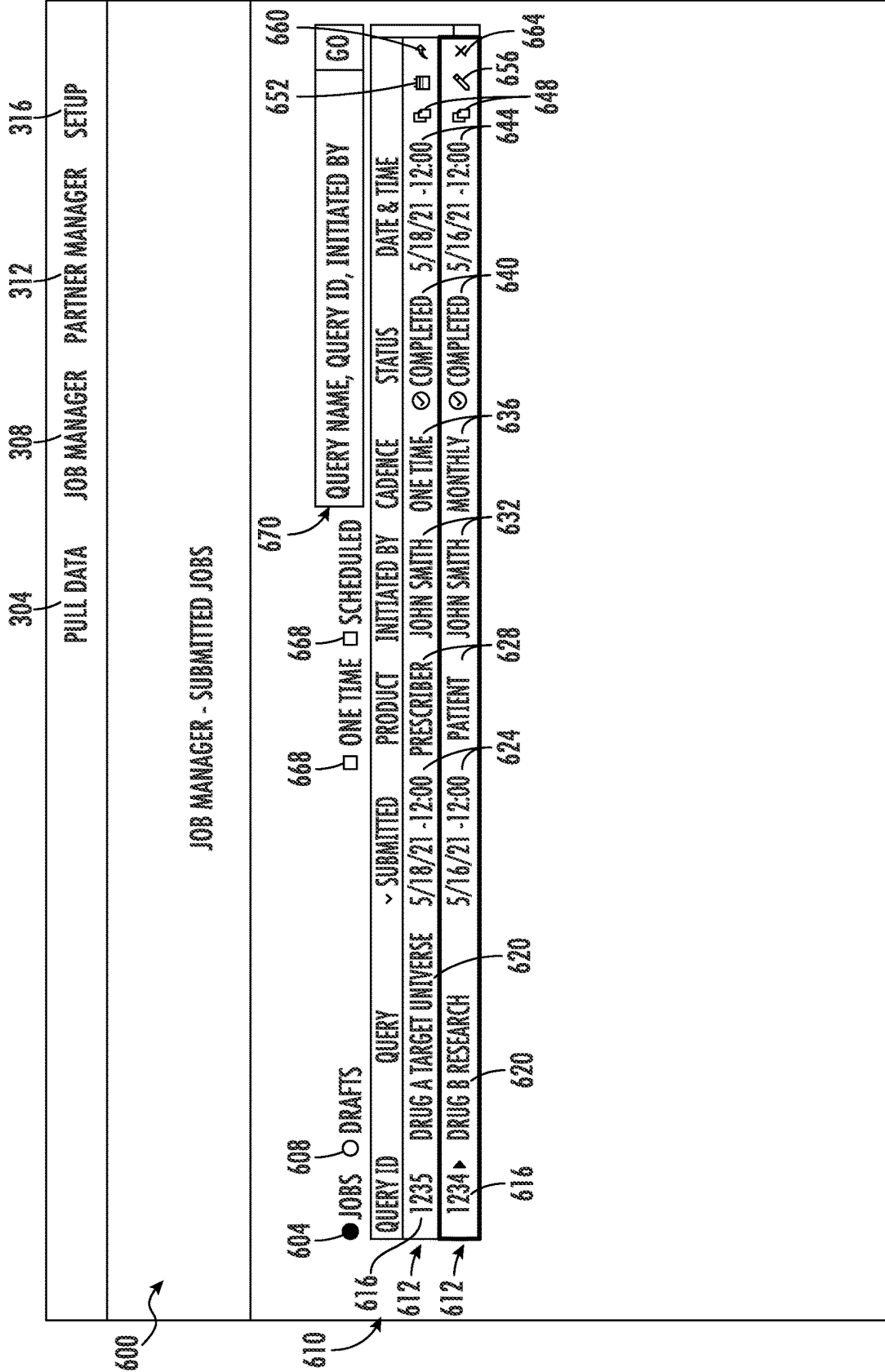


FIG. 6A

SHARE DATA - DRUG B ALL ACCESS

CONSULTING PARTNERS
SELECT YOUR PARTNER ▾

RECIPIENT(S)
ENTER RECIPIENT'S EMAIL

LICENSE
DRUG B ALL ACCESS

PROJECT NAME
ENTER CONSULTING PROJECT NAME

PROJECT DESCRIPTION
DESCRIBE PROJECT

PROJECT EXPIRATION DATE
MM/DD/YYYY

DATA SHARE REMINDER

SHARE CANCEL

JOBS	DRAFTS	QUERY ID	QUERY
●		1235	DRUG A TARGET UNIVERSE
		1234	DRUG B RESEARCH

১৬৬

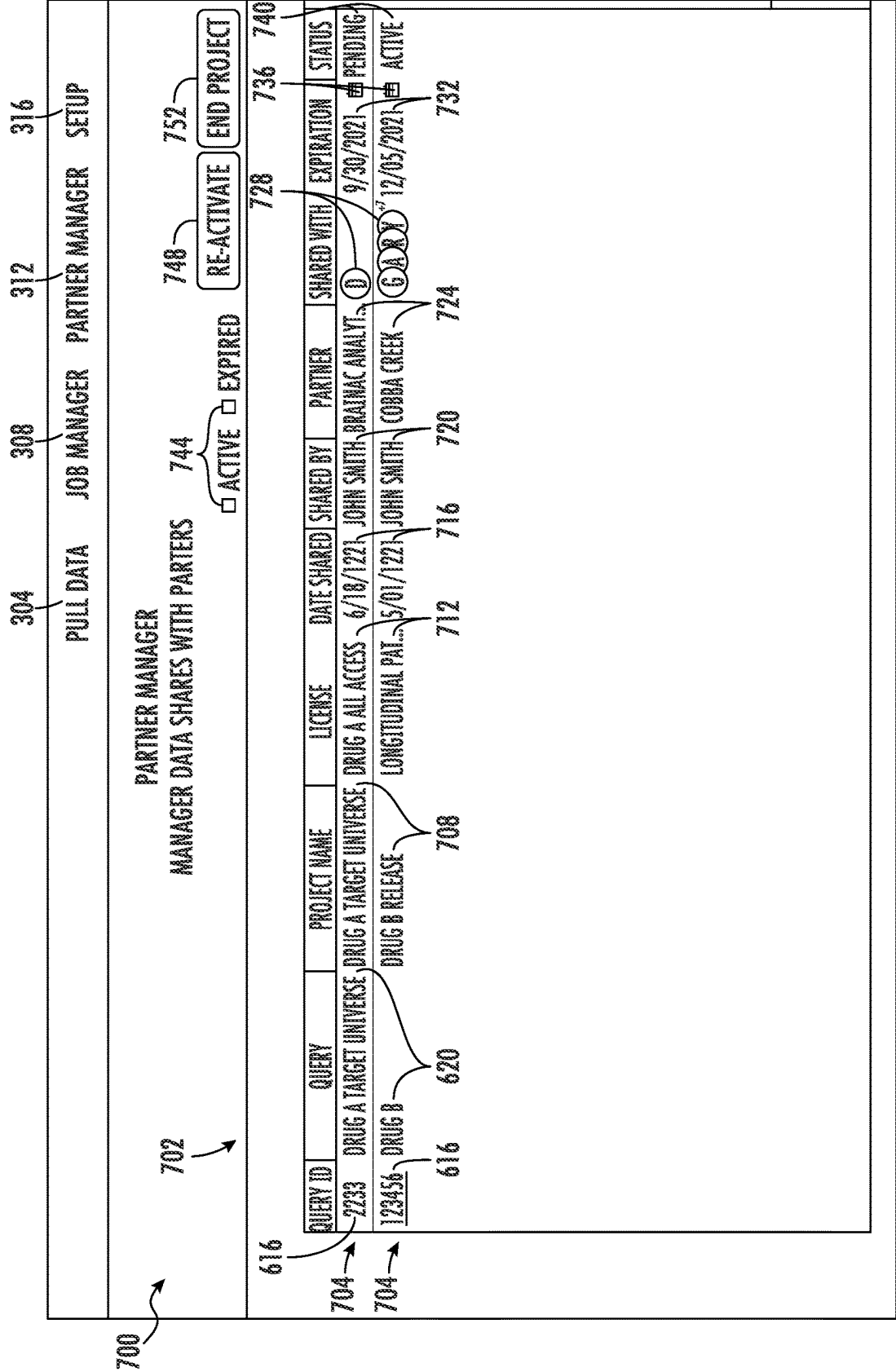


FIG. 7

NPI	VID	STATE	ZIP	SPECIALTY 1	SPECIALTY 2	HCP	NUMBER OF DIAGNOSES	DRUG 1 TRx	DRUG 2 TRx
11112222333	21112222333	STATE 1	11111	RHEUMATOLOGY	GENERAL PRACTICE	DOCTOR OF MEDICINE	66	1	25
11112222334	21112222334	STATE 2	11112	INTERNAL MEDICINE	RHEUMATOLOGY	DOCTOR OF MEDICINE	31		
11112222335	21112222335	STATE 3	11113	HOSPITALIST	RHEUMATOLOGY	DOCTOR OF MEDICINE	20		
11112222336	21112222336	STATE 4	11114	INTERNAL MEDICINE	ENDOCRINOLOGY	DOCTOR OF MEDICINE	62	1	5

FIG. 8A

800

DRUG 3 TRx	DRUG 4 TRx	DRUG 5 TRx	DRUG 6 TRx	DRUG 7 TRx	DRUG 8 TRx	DRUG 1 NBx	DRUG 2 NBx	DRUG 3 NBx	DRUG 4 NBx	DRUG 5 NBx	DRUG 6 NBx	DRUG 7 NBx	DRUG 8 NBx	# TOTAL PATIENTS	% MEDICARE	% MEDICAID
4						1	8	1						198	30	10
15	12							2	2					93	32	11
10					7			1					2	160	33	10
22	2				30	1	1	3	1				3	186	34	4

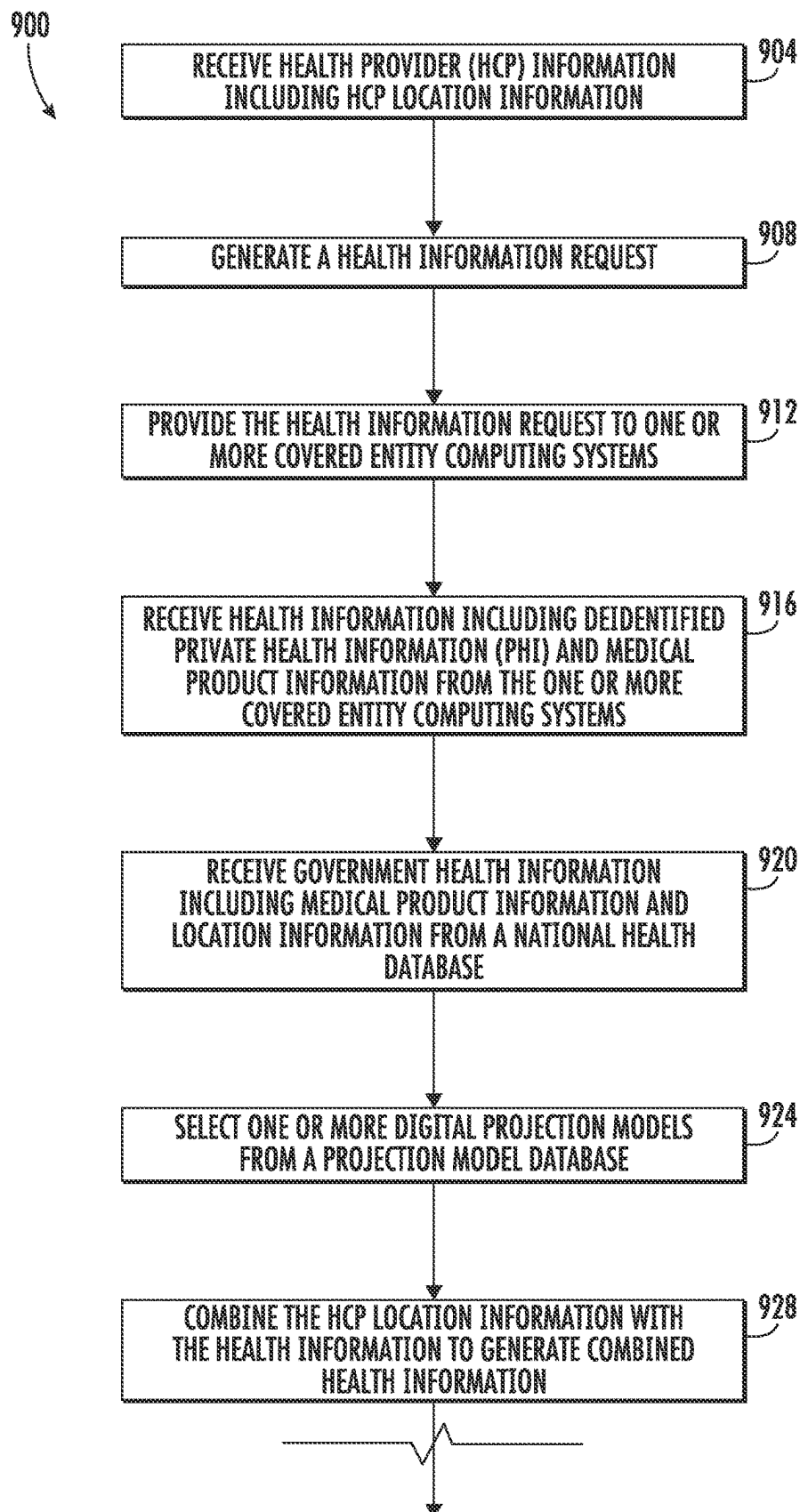
FIG. 8B

800

% COMMERCIAL	% CASH	% 0-10,000	% 10,001-40,000	% 40,001-90,000	% 90,001-120,000	% 120,001-200,000	% 200,001-500,000	% 500,000+
42	10	6	19	40	16	12	5	2
52	5	4	21	22	33	12	7	1
48	9	0	10	41	28	10	7	4
52	10	0	1	24	48	16	9	2

FIG. 8C

800



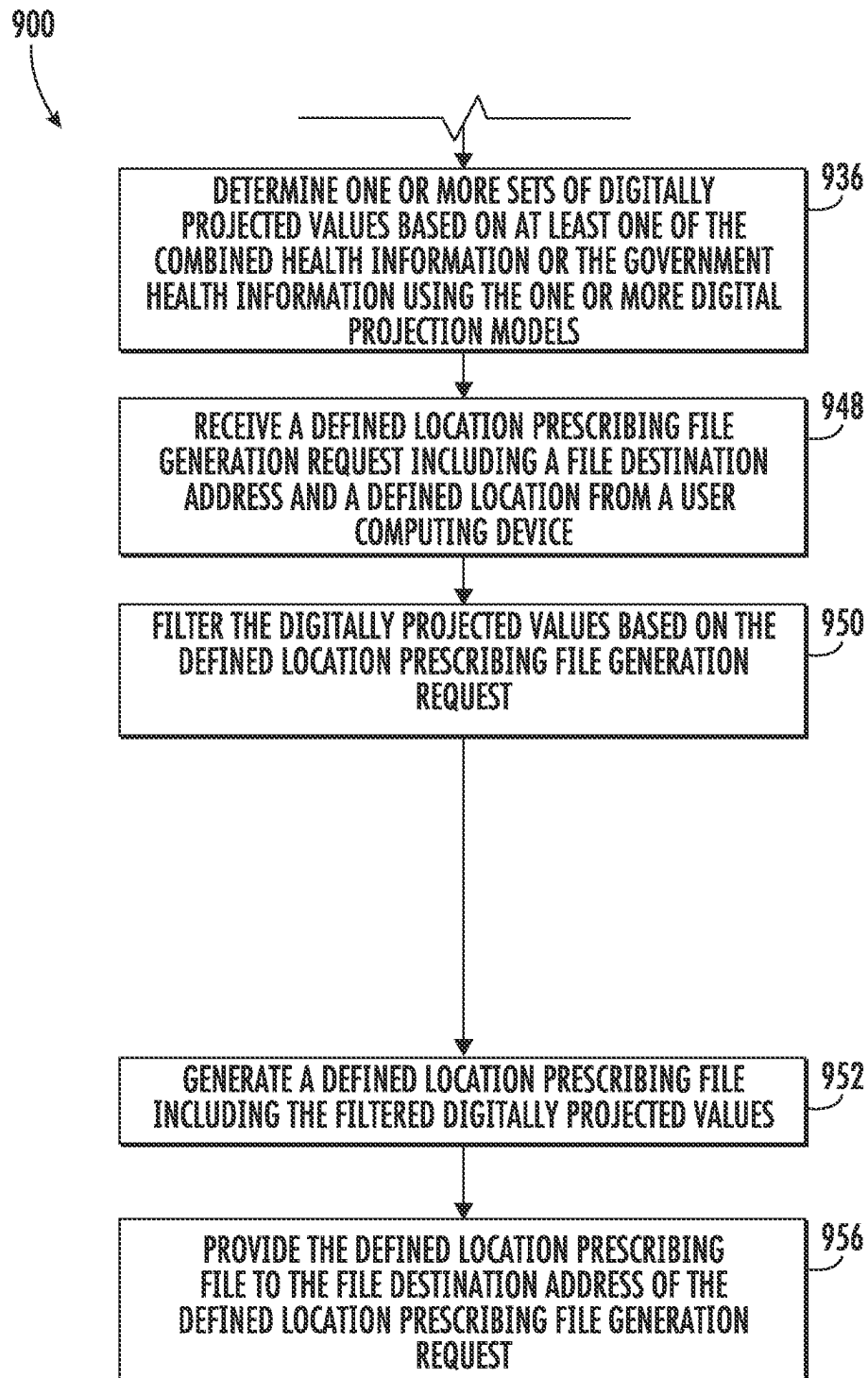


FIG. 9B

1004

QUERY: NOT SAVED

SAVE

SAVE AS

1008

SUBMIT

SALES DATA

1012

DEFINE MARKET

START WITH PRE-SET TEMPLATE(S) OR MANUALLY ADD PRODUCTS TO DEFINE A MARKET

1014

ADD PRODUCTS

1016

SELECT A TEMPLATE

1018

ADDED PRODUCTS - 32

REMOVE ALL

ALL RHEUMATOID ARTHRITIS TEMPLATE PRODUCT

ALL RHEUMATOID ARTHRITIS TEMPLATE PRODUCT 2

DRUG A

DRUG B

DRUG C

DRUG D

SAVE AS TEMPLATE

1022

ZIP CODES

ADD ZIP CODES ON STATE LEVEL

1024

ADD ALL

ALABAMA (21 ZIP CODES)

ALASKA (21 ZIP CODES)

ARIZONA (41 ZIP CODES)

ARKANSAS (21 ZIP CODES)

CALIFORNIA (243 ZIP CODES)

COLORADO (65 ZIP CODES)

CONNECTICUT (32 ZIP CODES)

DELAWARE (43 ZIP CODES)

FLORIDA (65 ZIP CODES)

1032

REMOVE ALL

FIG. 10A

1036

ATTRIBUTES

MEASUREMENTS

1040

TOTAL PRESCRIPTIONS (TRX)

1042

TOTAL MEDICAL CLAIMS (TMX)

1048

TIMEFRAME

1050

ROLLING TIME FRAME

1050

FIXED TIME FRAME

FROM:

1052

01/01/2020

1054

01/12/2021

1072

DELIVER OPTIONS

1076

DOWNLOAD TO MY LOCAL COMPUTER

1076

SEND TO DESTINATION

1076

CSV

1076

SFTP

1008

SUBMIT

1000

FIG. 10B

PRODUCT NAME	TIME_DIMENSION	TIME_VALUE	YEAR	DATE_ENDING	GEOGRAPHY	GEOGRAPHY DES MEASUREMENT	VALUE
DRUG A	MONTH	1	2021	2021-01-31	90640	ZIP5-90640	1
DRUG A	MONTH	1	2021	2021-01-31	680	ZIP5-00680	4
DRUG A	MONTH	1	2021	2021-01-31	8822	ZIP5-08822	10
DRUG A	MONTH	1	2021	2021-01-31	10013	ZIP5-10013	787
DRUG A	MONTH	1	2021	2021-01-31	32771	ZIP5-32771	1001
DRUG A	MONTH	1	2021	2021-01-31	31021	ZIP5-31021	54
DRUG A	MONTH	1	2021	2021-01-31	90505	ZIP5-90505	77
DRUG A	MONTH	1	2021	2021-01-31	94115	ZIP5-94115	23

1100

FIG. 11

SYSTEMS AND METHODS FOR ANALYZING LONGITUDINAL HEALTH INFORMATION AND GENERATING A DYNAMICALLY STRUCTURED ELECTRONIC FILE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a Continuation-In-Part of U.S. patent application Ser. No. 17/472,360, filed Sep. 10, 2021, which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

The present disclosure relates to systems and methods for analyzing longitudinal health information and generating a dynamically structured electronic file.

BACKGROUND

Researchers, scientists, industry players, academics, government regulators, and other stakeholders are increasingly in need of longitudinal health information and location-specific insights that are accessible, easy to interpret, analyze, and/or visualize.

SUMMARY

One embodiment relates to a method for generating a dynamically structured electronic file associated with a defined location. The method includes receiving government health information including location information from a national health database and generating a health information request. The method further includes providing the health information request to the one or more covered entity computing systems. The method further includes receiving health information including deidentified protected health information (PHI) from the one or more covered entity computing systems and health care provider (HCP) information including HCP location information. The method further includes combining the health information with the HCP location information to generate combined health information. The method further includes selecting one or more digital projection models from a projection model database. The method further includes determining multiple digitally projected values based on at least one of the combined health information or the government health information using at least one of the one or more digital projection models. Each digitally projected value of the plurality of digitally projected values is associated with a specific location. The method further includes receiving a file generation request from the user computing device. The file generation request identifying a file and including a file destination address and a defined location. The method further includes filtering the multiple digitally projected values for the defined location of the file generation request by removing digitally projected values that are not associated with the defined location. The method further includes generating the electronic file including the filtered digitally projected values and at least a portion of the location information or the HCP location information. The electronic file is generated as the file type identified by the file generation request. The electronic file is structured based on at least one of the file generation request or the file destination. The method further includes providing the electronic file to the file destination address of the file generation request.

Another embodiment relates to a method for generating a dynamically structured electronic file associated with a defined location. The method includes receiving government health information including location information from a national health database. The method further includes receiving health information including deidentified protected health information (PHI) from the one or more covered entity computing systems and health care provider (HCP) information including HCP location information. The method further includes combining the health information with the HCP location information to generate combined health information. The method further includes selecting one or more digital projection models from a projection model database. The method further includes determining a first set of digitally projected values based on the government health information using the one or more digital projection models and a second set of digitally projected values based on the combined health information and the government health information using at least one of the one or more digital projection models. Each digitally projected value of the first set of digitally projected values and the second set of digitally projected values is associated with a specific location. The method further includes receiving a file generation request from the user computing device. The file generation request identifying a file and including a file destination address and a defined location. The method further includes filtering the first set and second set of digitally projected values for the defined location of the file generation request by removing digitally projected values that are not associated with the defined location. The method further includes generating the electronic file including the filtered digitally projected values and at least a portion of the location information or the HCP location information. The electronic file is generated as the file type identified by the file generation request. The electronic file is structured based on at least one of the file generation request or the file destination. The method further includes providing the electronic file to the file destination address of the file generation request.

This summary is illustrative only and is not intended to be in any way limiting. Other aspects, inventive features, and advantages of the devices or processes described herein will become apparent in the detailed description set forth herein, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 is a component diagram of a longitudinal health information analyzation and dynamically structured electronic file generation system, according to an example embodiment.

FIGS. 2A and 2B are a flow diagram of a method for analyzing longitudinal health information and generating and sharing a dynamically structured electronic HCP file, according to an example embodiment.

FIG. 3 is an illustration of some aspects of a user interface generated by the longitudinal health information analyzation and dynamically structured electronic file generation system of FIG. 1 to determine HCP information is sought, according to an example embodiment.

FIGS. 4A-4C are multiple illustrations of some aspects of a user interface generated by the longitudinal health information analyzation and dynamically structured electronic file generation system of FIG. 1 to determine the structure and file type of the electronic HCP file, according to an example embodiment.

FIG. 5 is an illustration of some aspects of a user interface generated by the longitudinal health information analyzation and dynamically structured electronic file generation system of FIG. 1 to modify a medical product group, according to an example embodiment.

FIG. 6A-6B are multiple illustrations of some aspects of a user interface generated by the longitudinal health information analyzation and dynamically structured electronic file generation system of FIG. 1 to manage HCP file generation requests, according to an example embodiment.

FIG. 7 is an illustration of some aspects of a user interface generated by the longitudinal health information analyzation and dynamically structured electronic file generation system of FIG. 1 to manage sharing of a generated electronic HCP file.

FIGS. 8A-8C are an illustration of some aspects of an electronic HCP file generated by the longitudinal health information analyzation and dynamically structured electronic file generation system of FIG. 1, according to an example embodiment.

FIGS. 9A-9B are a flow diagram of a method for analyzing longitudinal health information and generating a dynamically structured electronic defined location prescribing file, according to an example embodiment.

FIGS. 10A-10B are multiple illustrations of some aspects of a user interface generated by the longitudinal health information analyzation and dynamically structured electronic file generation system of FIG. 1 to receive a defined location prescribing file generation request, according to an example embodiment.

FIG. 11 is an illustration of some aspects of an electronic defined location prescribing file generated by the longitudinal health information analyzation and dynamically structured electronic file generation system of FIG. 1, according to an example embodiment.

DETAILED DESCRIPTION

Referring generally to the figures, systems and methods for analyzing longitudinal health information and generating a dynamically structured electronic HCP file are disclosed. The systems and methods described herein provide for enhanced analyzation of longitudinal health information and generation HCP insights by utilizing a wide variety of longitudinal health information, HCP information, and consumer information. For example, because the systems and methods described utilize longitudinal health information including consistently deidentified private health information (PHI), the systems described herein can make associations across all the received information while protecting the patients' identities based solely on the deidentified PHI of the health information, which reduces the complexity of the analysis of the longitudinal health information, HCP information, and consumer information and therefore requires less processing power by the systems described herein. For example, instead of having to associate separately received health information based on multiple matching attributes (e.g., zip code, type of medical coverage, cost of medical procedure, etc.), the systems described herein can associate the separately received health information based solely on the deidentified PHI of the health information, due to the consistent deidentification of the PHI.

The systems and methods described herein further provide for improved structuring and generation of electronic files (also referred herein as HCP files or defined location prescribing files). For example, because the electronic file is dynamically structured and includes specific HCP informa-

tion, health information, and HCP insights, as decided by a user, the electronic file does not include excess or unwanted values and therefore requires less memory to be stored and less processing power to be shared between computing devices and systems. For example, if the electronic file were generated as a statically structured file (e.g., included the same values every time) and still included all the values, insights, and information described herein, the electronic file would include excess information that was unwanted by the user or the computing device/system to which the electronic file is provided. The excess information would make the electronic file larger, thereby requiring additional memory to store the electronic file and additional processing power to generate and share the electronic file. In comparison, because the systems and methods described herein dynamically structure the electronic file, as decided by the user and/or the device/system to which the electronic file is provided, the electronic file includes only requested information and HCP insights, reducing the memory required by the provider computing system to store the electronic file and processing power required by the provider computing system to generate and share the electronic file.

Likewise, because the electronic file is dynamically generated as a specified file type, as decided by the user or the computing device/system to which the electronic file is provided, the systems and methods described herein may provide for improved shareability of the electronic file as well as use less processing power and memory. For example, in situations where the electronic file is only generated as a static file type, third-party computing systems and recipient computing devices (e.g., the recipient computing device) may have to perform additional file modification and conversion or request the provider computing system to do so, which requires additional processing power, memory (to keep two files), and time. In comparison, the systems and methods described herein allow the user to specify the file type of the electronic file through the electronic file generation request. Accordingly, the user can specify the wanted file type the first time and the provider computing system will generate the electronic file as the specified file type. As a result, no additional file modification or conversion is required reducing processing power usage, memory usage, and overall processing time for the provider computing system.

In an illustrative scenario, the provider computing system 104 may generate a health information request and provide the health information request to one or more covered entity computing systems. The covered entity computing systems, in response to the health information request, may determine health information matching the health information request, consistently deidentify PHI of the health information, and provide the health information, including the deidentified PHI to the provider computing system. The provider computing system may then receive an electronic file generation request from a user computing device and match the HCP information with the received health information. The provider computing system may further determine one or more HCP statistical values or insights based on the matched electronic file generation request, the matched HCP information, and the matched health information. The provider computing system may then generate an electronic file including the matched HCP information, the matched health information, and the HCP statistical values based solely on the HCP file generation request and as a specific file type based on the electronic file generation request.

The provider computing system may store the electronic file in a database as well as provide the electronic file to a

5

file destination. Lastly, the provider computing system may receive a file share request including a file share expiration date/time and share the file with a recipient, as indicated by the user computing device, until the file share expiration date/time is met and/or passed.

As described herein, “health information” can include information related to a patient’s medical history such as diagnosis information (e.g., an ICD-10 diagnosis code), procedures (e.g. a HCPCS procedure code), medical claims (Mx) information, clinical information, electronic health record information, government health information (e.g., national health insurance (e.g., Medicare, Medicaid, national/universal health care) claims information), facility or health care system location information, and medical outcomes as well as (pharmacy-dispensed) prescription claims (Rx) information associated with one or more medical products (e.g., including transaction information, prescribing HCP national prescriber identifier (NPI), medical product information, date/time information, diagnosis information, and the like). Further, “health information” can further include protected health information (PHI) such as a patient’s name, address, IP address, contact information, and the like and longitudinal health information. In this regard, “longitudinal health information” can include health information for a specific HCP or patient over a period of time. For example, “longitudinal health information” may include a group or set of Rx information for one patient over 10 years. In another example, “longitudinal health information” may include a group of Rx information for one HCP over a period of 10 years.

As described herein, “health care professional (HCP) information” can include information related to an HCP such as an NPI of the HCP, a name of the HCP, an address of the HCP, a specialty of the HCP (e.g., Cardiology), a place of employment of the HCP (e.g., Johns Hopkin’s hospital, X Medical Group, an address including zip code and/country, etc.), HCP location information (e.g., an address including zip code and/or country, a location of the HCP’s employer or primary health care organization (HCO), a country of residence or employment of the HCP, etc.) associated with the HCP, a type, degree, or certification held by the HCP (e.g., Doctor of Medicine, Registered Nurse, Nurse Practitioner, etc.), and other information described herein as being associated or related to an HCP.

Referring now to FIG. 1, a system 100 for analyzing longitudinal health information and generating a dynamically structured electronic file is shown, according to an example embodiment. The system 100 includes a provider computing system 104, a covered entity computing system 108, a user computing device 112, a recipient computing device (or system) 116, a consumer database 120, a health care professional (HCP) database 124, and one or more national health databases 125 connected by a secure network (e.g., a network 118).

The network 118 communicably and operably couples the provider computing system 104, the covered entity computing system 108, the user computing device 112, and the recipient computing device 116 such that communicable and operable computing may be provided between the provider computing system 104, the covered entity computing system 108, the user computing device 112, and the recipient computing device 116 over the network 118. In some embodiments, the network 118 further communicably and operably couples the consumer database 120, the HCP database 124, and the national health databases 125 to the provider computing system 104, the covered entity computing system 108, the user computing device 112, and the

6

recipient computing device 116. In various embodiments, the network 118 includes any combination of a local area network (LAN), an intranet, the Internet, or any other suitable communications network, directly or through another interface.

The provider computing system 104 may be operated and managed by a provider (e.g., a software as a service (SaaS) provider, a cloud services provider, a software provider, a service provider, etc.) and may include a computer system (e.g., one or more servers (e.g., a cloud computing server) each with one or more processing circuits). In some embodiments, the provider computing system 104 may act as a host and provide an application (e.g., a web-based application, a mobile application, etc.) to the user computing device 112 over the network 118 in response to authenticating the user computing device 112. As shown, the provider computing system 104 may include a network interface circuit 126, a processing circuit 128, an account database 132, a medical domain database 134, and a projection model database 135. In some embodiments, the provider computing system 104 may include an input/output circuit (e.g., similar to or the same as an input/output circuit 180 which will be described further herein).

The network interface circuit 126 is structured to establish connections with the covered entity computing system 108, the user computing device 112, the recipient computing device 116, the consumer database 120, and the HCP database 124 by way of the network 118. The network interface circuit 126 includes program logic and/or hardware-based components that connect the provider computing system 104 to the network 118. For example, the network interface circuit 126 may include any combination of a wireless network transceiver (e.g., a cellular modem, a broadband modem, a Bluetooth transceiver, a Wi-Fi transceiver, a Li-Fi transceiver, etc.) and/or a wired network transceiver (e.g., an Ethernet transceiver). In some embodiments, the network interface circuit 126 includes the hardware and machine-readable media structured to support communication over multiple channels of data communication (e.g., wireless, Bluetooth, near-field communication (NFC)). In some embodiments, the network interface circuit 126 includes cryptography logic and capabilities to establish a secure communications session.

The processing circuit 128, as shown, comprises a memory 136, a processor 140, a request generation and management circuit 144, an information matching and analytics circuit 148, a file generation circuit 150, and a file sharing circuit 152. The memory 136 includes one or more memory devices (e.g., RAM, NVRAM, ROM, flash memory, hard disk storage, etc.) that store data and/or computer code for facilitating the various processes described herein. That is, in operation and use, the memory 136 stores at least portions of instructions and data for execution by the processor 140 to control the processing circuit 128. The memory 136 may be or include tangible, non-transient volatile memory and/or non-volatile memory. The processor 140 may be implemented as a general-purpose processor, an application specific integrated circuit (ASIC), one or more field programmable gate array (FPGAs), a digital signal processor (DSP), a group of processing components or other suitable electronic processing components.

As described herein, the request generation and management circuit 144 is structured to generate and manage health information requests that may be provided to the covered entity computing system 108. For instance, the request generation and management circuit 144 may be structured to

retrieve information included in the health information request (e.g., a timeframe for which health information is sought), government health information (e.g., from the national health databases **125**), consumer information (e.g., from the consumer database **120**) for which matching or associated health information is sought, HCP information (e.g., from the HCP database **124**) for which matching or associated health information is sought, medical product information for which matching or associated health information is sought (e.g., from the medical domain database **134**), or any combination thereof. Further, the request generation and management circuit **144** may be structured to generate the health information request including the information included in the health information request and provide the health information request to the covered entity computing system **108**. In some embodiments, the request generation and management circuit **144** may be configured to generate and provide the health information request to the covered entity computing system **108** at specific time intervals (e.g., every hour, every day, every week, every month, etc.), and, as a result, the provider computing system **104** may receive the health information at the specific time intervals. By doing so, the provider computing system **104** may act as a centralized source of reliable, up to date, health information. For instance, because the request generation circuit **144** may generate and provide the health information request to the covered entity computing system **108** at specific time intervals, the provider computing system **104** may receive updated health information at the specific time intervals and have the most up to date health information.

As described herein, the health information request may include information for which matching health information is sought. In one example, the health information request may include a list of specific IP addresses that watched an online advertisement. The covered entity computing system **108** may receive the health information request, match (longitudinal) health information of patients having an IP address within the list and return the health information to the provider computing system **104**. Prior to returning the matching health information, the covered entity computing system **108** may de-identify (e.g., hash, tokenize, encrypt, etc.) the PHI included in the health information. In this way, the PHI of the patient is protected, and the patient's privacy is maintained.

In some embodiments the request generation and management circuit **144** is further structured to manage and receive health care professional (HCP) file generation requests that may be received from the user computing device **112**. For instance, the request generation and management circuit **144** may receive one or more HCP file generation requests (also referred to as electronic file generation requests) from the user computing device **112**. The HCP file generation requests may each indicate one or more statistical values (e.g., median quantity of a medical product, likelihood or percentage of times switched from the medical product, percentage of generic medical products versus name brand medical products, etc.) associated with the HCP that are to be calculated as well as a file destination address to which the HCP file may be provided. In other embodiments, a separate request circuit or other circuits (e.g., the information matching and analytics circuit **148**) described herein may manage and receive HCP file generation requests from the user computing device **112**.

In some embodiments the request generation and management circuit **144** is further structured to manage a defined location prescribing file generation request (also referred to as an electronic file generation request) that may be received

from the user computing device **112**. For instance, the request generation and management circuit **144** may receive one or more defined location prescribing file generation request from the user computing device **112**. The defined location prescribing file generation requests may each indicate one or more defined locations (e.g., one or more zip codes, one or more states, one or more countries) for which projected prescribing values are sought and statistical values (e.g., median quantity of a medical product, likelihood or percentage of times switched from the medical product, percentage of generic medical products versus name brand medical products, etc.) associated with the defined location that are to be calculated. Further, the defined location prescribing file generation request may include a file destination address to which the defined location prescribing file may be provided. In other embodiments, a separate request circuit or other circuits (e.g., the information matching and analytics circuit **148**) described herein may manage defined location prescribing file generation requests.

The information matching and analytics circuit **148** is structured to receive the health information from the covered entity computing system **108** and/or government health information from the national health databases **125**, match the health information and/or government health information with HCP information (e.g., the location information of the HCP information) from the HCP database **124**, filter the matching health information and HCP information (e.g., as specified by the HCP file generation request), and determine one or more HCP statistical values or insights based on the matched and filtered health information and HCP information. For example, the provider computing system **104** may receive an HCP file generation request that requests health information associated with all HCPs who prescribed a patient drug X to be included in the HCP file. Moreover, the HCP file generation request may also request the provider computing system **104** to include the median quantity of drug X that each HCP prescribes. The information matching and analytics circuit **148** may match the health information with the prescriber information (e.g., based on the NPI of the HCP in both sets of information) and filter the matched health information for medical product drug X (i.e., remove all matched health information and HCP information that does not include medical product information of drug X). Further, the information matching and analytics circuit **148** may calculate the median quantity of drug X for each HCP who prescribed drug X using the filtered and matched health information. In this way, the information matching and analytics circuit **148** may generate statistical values or insights for each HCP based on the longitudinal health information matched with the HCP.

In another example, the provider computing system **104** may receive a defined location prescribing file generation request that requests health information associated with each zip code of a first state (e.g., Wisconsin) and drug X, for the past year, to be included in the defined location prescribing file. Moreover, the defined location prescribing file generation request may also request the provider computing system **104** to include the median quantity of drug X that was prescribed in each zip code of the state. The information matching and analytics circuit **148** may match the health information with the HCP information (e.g., based on the NPI of the HCP in both sets of information) and filter the matched health information for medical product drug X and the defined location (i.e., remove all matched health information and HCP information that does not include medical product information of drug X and location information that matches Wisconsin). Further, the information matching and

analytics circuit **148** may calculate the median quantity of drug X for each HCP who prescribed drug X using the filtered and matched health information.

In some embodiments, the information matching and analytics circuit **148** is further structured to determine if the health information received from the covered entity computing system **108** includes duplicate information. Because the health information can be received from a variety of covered entity computing systems, duplicate information is common and needs to be accounted for to provide for accurate insights. As a result, the information matching and analytics circuit **148** may parse the received health information and remove (if exact duplicates) or combine (if partial duplicates) sets of duplicate health information. By doing so, the provider computing system **104** may act as a trusted health information repository and maintain information integrity.

In some embodiments, the information matching and analytics circuit **148** is structured to select one or more digital projection models from the projection model database **135** and determine one or more sets of digitally projected prescriber values based on the matched government health information, the matched HCP information (e.g., location information), and/or the matched health information. For instance, the information matching and analytics circuit **148** may select a first digital projection model from the digital projection model database **135** and determine a first set of digitally projected prescriber values based on the government health information and a second set of digitally projected prescriber values based on the government health information, the matched/combined health information, and the matched/combined location information. In some embodiments, a separate modeling circuit (e.g., not shown) may select the one or more digital projection models and determine the one or more sets of digitally projected prescriber values.

The file generation circuit **150** is structured to generate the HCP file based on the HCP file generation request received by the provider computing system **104**. For instance, once the information matching analytics circuit **144** has matched the health information with the HCP information and generated any requested statistical values, the file generation circuit **150** may generate a file of a specified format (e.g., comma-separated value (CSV) format, Excel (XLS) format, etc.) and populate it with the matched health information, the matched HCP information, and the generated statistical values. Because the HCP file is generated as a specified file format, the HCP file can be shared with a variety of destinations and quickly parsed and analyzed. In some embodiments, the HCP file generation request may indicate the specified file format that the HCP file is to be.

The file sharing circuit **152** is structured to share and manage the HCP file with computing devices, customers, and user's other than the user of the user computing device **112** (e.g., recipients). For instance, the file sharing circuit **152** may receive a file share request indicating a recipient address and a file share expiration date and/or time from the user computing device **112**. In response, the file sharing circuit **152** may share the HCP file with the recipient address and/or allow the recipient address to access the HCP file until the expiration date and/or time is met. In this way, the user of the user computing device **112** may easily provide others access to the HCP file and share the HCP file for improved collaboration and analysis of the HCP insights.

The account database **132** is a repository that receives, stores, and manages various account information (e.g., login credentials, HCP files, recipient devices to share HCP files

with, etc.). To do so, the account database **132** can be structured according to various database types, such as, relational, hierarchical, network, flat, point-in time, and/or object relational. In operation, the account database **132** is configured to store account information in association with a specific user or specific users. For instance, the account database **132** may store generated HCPs file in association with the user of the user computing device **112**. In one embodiment, the account database **132** may store medical product groups (i.e., multiple associated medical products) associated with the account of the user. In other embodiments, the medical domain database **134**, as described herein, may do so. In some embodiments, multiple databases may take the place of a single account database and store/manage one or more pieces of the information that the account database is described herein as storing/managing.

Likewise, the medical domain database **134** is a repository that receives, stores, and manages various medical domain information (e.g., medical product information, procedure information, diagnosis information (e.g., diagnosis codes and descriptions), and procedure information (e.g., procedure codes and descriptions) associated with the user or an HCP. To do so, the medical domain database **134** can be structured according to various database types, such as, relational, hierarchical, network, flat, point-in time, and/or object relational. In operation, the medical domain database **134** is configured to store medical domain information such as medical product information, diagnosis information, and procedure information. In one instance, the medical domain database **134** may receive a grouping of multiple drugs as specified by the user of the user computing device **112**. As will be described herein, the grouping of drugs may be utilized by the user in generating HCP file generation requests and HCP files.

Similarly, the projection model database **135** is a repository that receives, stores, and manages various digital projection models (e.g., machine-learning (ML) models, statistical models, etc.) used to determine digitally projected prescription values. To do so, the projection model database **135** can be structured according to various database types, such as, relational, hierarchical, network, flat, point-in time, and/or object relational. In operation, the projection model database **135** is configured to store, manage, and provide digital projections models such as ML projection (i.e., forecasting) models (e.g., binary classification models, neural network models (e.g., Convolutional Neural Network (CNN) model, Long Short-Term Memory Networks (LSTMs) models) multiclass classification models, regression models (e.g., linear regression models), etc.).

In some embodiments, the provider computing system **104** may further include a digitally projected value database (not shown). The digitally projected value database may receive, store, and manage the generated digitally projected values described further here. The digitally projected value database can be structured according to various database types, such as, relational, hierarchical, network, flat, point-in time, and/or object relational. In operation, the digitally projected value database may receive digitally projected values and associated metadata. For instance, the digitally projected value database may receive a digitally projected value and metadata that indicates the digitally projected value is associated with a specific timeframe, a specific medical product, a specific prescription type, and/or a specific location. Then, the provider computing system **104** may select or query the digitally projected value database for digitally projected values based on the metadata.

11

Still referring to FIG. 1, the covered entity computing system 108 may include a computer system (e.g., one or more servers each with one or more processing circuits) and be operated by, be managed by, and/or operate on a data network of a “covered entity” as defined by the Health Insurance Portability and Accountability Act of 1996 (HIPAA) (or other laws that protect health information). For example, the covered entity may be a pharmacy chain (e.g., a pharmacy company) with access to prescription claims information, a medical claims company (e.g., a medical insurance company) with access to medical claims information, a medical provider with access to electronic health records, or a government entity with access to national health insurance (e.g., Medicare or Medicaid) claims information, and the like. In this regard, the covered entity computing system 108 may have direct access to health information including PHI, transaction information, and other health information associated with one or more patients. As shown, the covered entity computing system 108 may include a network interface circuit 156, a processing circuit 160, and a prescription and medical claims database 164. In some embodiments, the covered entity computing system 108 may include an input/output circuit (e.g., similar to the input/output circuit 180).

The network interface circuit 156 is structured to establish connections with the provider computing system 104, the user computing device 112, the recipient computing device 116, the consumer database 120, and the HCP database 124 by way of the network 118. The network interface circuit 156 includes program logic and/or hardware-based components that connect the covered entity computing system 108 to the network 118. For example, the network interface circuit 156 may include any combination of a wireless network transceiver (e.g., a cellular modem, a broadband modem, a Bluetooth transceiver, a Wi-Fi transceiver, a Li-Fi transceiver, etc.) and/or a wired network transceiver (e.g., an Ethernet transceiver). In some embodiments, the network interface circuit 156 includes the hardware and machine-readable media structured to support communication over multiple channels of data communication (e.g., wireless, Bluetooth, near-field communication (NFC)). In some embodiments, the network interface circuit 156 includes cryptography logic and capabilities to establish a secure communications session.

The processing circuit 160, as shown, comprises a memory 168, a processor 170, an information matching circuit 172, and a deidentification circuit 174. The memory 168 includes one or more memory devices (e.g., RAM, NVRAM, ROM, flash memory, hard disk storage, etc.) that store data and/or computer code for facilitating the various processes described herein. That is, in operation and use, the memory 168 stores at least portions of instructions and data for execution by the processor 170 to control the processing circuit 160. The memory 168 may be or include tangible, non-transient volatile memory and/or non-volatile memory. The processor 170 may be implemented as a general-purpose processor, an application specific integrated circuit (ASIC), one or more field programmable gate array (FPGAs), a digital signal processor (DSP), a group of processing components or other suitable electronic processing components.

The information matching circuit 172 is structured to receive the health information requests from the provider computing system 104 including any additional information for which matching health information is sought (e.g., consumer information, HCP information, etc.) and determine the matching health information. For instance, the

12

information matching circuit 172 may receive a health information request including medical product information (e.g., a list of medical products) for which matching health information is sought. The information matching circuit 172 may then search or query the prescription and medical claims database 164 for health information that is associated with one or more of the medical products of the medical product information. For example, the information matching circuit 172 may search and return a piece of health information (e.g., an Rx transaction) which is associated with a medical product of the medical product information. In another example, the information matching circuit 172 may search and return health information matched with consumer information (e.g., demographic information).

The deidentification circuit 174 is structured to receive the matched health information and deidentify any PHI included within the matched health information and corresponding information (e.g., consumer information). To deidentify the PHI of the matched health information, the deidentification circuit 174 may first determine any PHI included in the matched health information and then perform one or more operations to remove the associated patient's identity or related private information. In some embodiments, the deidentification circuit 174 may deidentify PHI of the health information by encrypting the PHI, tokenizing the PHI, or by performing some other information masking function.

In other embodiments, the deidentification circuit 174 may deidentify PHI of the health information by performing a hash function on the PHI. For example, the deidentification circuit 174 may perform a hash function on the private health information that removes the identity or private information of the patient and returns a hash value in place of the PHI. Example hash functions include, but are not limited to, the mid-square method, string folding, and the like. In some embodiments, all covered entity computing systems from which health information is received by the provider computing system 104 (including the covered entity computing system 108) may use the same hash function to provide for consistent hash values across covered entity computing systems. For example, because each covered entity computing system may use the same hash function (i.e., a hash function that returns the same hash value for the same input), the provider computing system 104 may receive health information, with deidentified PHI, from multiple covered entity computing systems. For example, the provider computing system 104 may receive health information including Rx information from a pharmacy chain covered entity computing system, health information including Mx information from a medical claims company, health information including clinical information from a medical provider (e.g., a hospital), and health information including national health insurance claims information from a national health insurance provider, each piece of health information including deidentified PHI. Then, the provider computing system 104 may match the various pieces of received health information based on the deidentified PHI and specifically based on matching hash values. In other embodiments, the provider computing system 104 matches the various pieces of received health information by matching one or more patient demographics or attributes (e.g., same HCP NPI, same date of transaction, same credit card used in both transactions, etc.) to match the health information. In this way, information integrity and usability of the PHI may be maintained while protecting the patients PHI and identity.

Once the deidentification circuit 174 has deidentified the PHI of the health information, the processing circuit 160 and

13

the network interface circuit **156** may be structured to send the health information, including the deidentified PHI, to the provider computing system **104**, in response to the health information request. In some embodiments, the covered entity computing system **108** may receive health information requests at specific time intervals and provide the matched health information to the provider computing system **104** at the same time intervals. In this way, the provider computing system **104** may have the most up to date health information and act as a centralized source of reliable, up to date, health information.

The prescription and medical claims database **164** is a repository that receives, stores, and manages various Rx information and/or Mx information associated with the customers or users of the covered entity. To do so, the prescription and medical claims database **164** can be structured according to various database types, such as, relational, hierarchical, network, flat, point-in time, and/or object relational. In operation, the prescription and medical claims database **164** is configured to store Rx and/or Mx information. In some embodiments, the prescription and medical claims database **164** is a health information database **164** configured to store health information including PHI, Rx information, medical product information, Mx information, and the like. For example, if the covered entity is a medical claims company, the covered entity computing system may include a medical claims database to store medical claims information. In another example, the covered entity computing system **108** may include a national health insurance claims database (e.g., similar to or the same as the national health database **125**) or an electronic health record database. In some embodiments, the prescription and medical claims database **164** is not included in the covered entity computing system **108** but is connected to the covered entity computing system **108** over a covered entity private network or the network **118**.

In other embodiments, each covered entity computing system **108** may include multiple databases information databases for each piece of separately stored health information (i.e., a separate database for Rx information and for Mx information).

Still referring to FIG. 1, the user computing device **112** can be any type of computing device or system configured to run an application (e.g., a web-based application, a mobile application, etc.). For instance, the user computing device **112** can be one or more of a mobile phone, a tablet computer, a laptop computer, a smart watch, a server computer system, and any other internet-connected device that is capable of running an application. In operation, the user computing device **112** may download or connect to (e.g., via the network **118**) the application of the provider computing system **104**. Once downloaded or connected to the application of the provider computing system **104**, the user computing device **112** may interface with the provider computing system **104** to receive the HCP file, to generate and provide an HCP file generation request, and to generate and provide an HCP file share request. As shown, the user computing device **112** includes a network interface circuit **176**, a processing circuit **178**, and the input/output (I/O) circuit **180**.

The network interface circuit **176** is structured to establish connections with the provider computing system **104**, the covered entity computing system **108**, the recipient computing device **116**, the consumer database **120**, and the HCP database **124** by way of the network **118**. The network interface circuit **176** may be similar to the network interface circuit **126** and include program logic and/or hardware-

14

based components that connect the user computing device **112** to the network **118**. For example, the network interface circuit **176** may include any combination of a wireless network transceiver (e.g., a cellular modem, a broadband modem, a Bluetooth transceiver, a Wi-Fi transceiver, a Li-Fi transceiver, etc.) and/or a wired network transceiver (e.g., an Ethernet transceiver). In some embodiments, the network interface circuit **176** includes the hardware and machine-readable media structured to support communication over multiple channels of data communication (e.g., wireless, Bluetooth, near-field communication (NFC)). In some embodiments, the network interface circuit **176** includes cryptography logic and capabilities to establish a secure communications session.

The processing circuit **178**, as shown, may comprise a memory **182**, a processor **184**, and a user interface generation circuit **185**. The memory **182** includes one or more memory devices (e.g., RAM, NVRAM, ROM, flash memory, hard disk storage, etc.) that store data and/or computer code for facilitating the various processes described herein. That is, in operation and use, the memory **182** stores at least portions of instructions and data for execution by the processor **184** to control the processing circuit **178**. The memory **182** may be or include tangible, non-transient volatile memory and/or non-volatile memory. The processor **184** may be implemented as a general-purpose processor, an application specific integrated circuit (ASIC), one or more field programmable gate array (FPGAs), a digital signal processor (DSP), a group of processing components or other suitable electronic processing components.

The user interface generation circuit **185** may be configured to receive a user interface (e.g., a web interface in an HTML file and related files, a downloaded graphical user interface, etc.) from the provider computing system **104** and render or generate the user interface on the user computing device **112** via the I/O circuit **180**. In this way, the provider computing system **104** may generate one or more user interfaces and provide the one or more user interfaces to the user interface generation circuit **185** to be rendered on the user computing device **112** (e.g., on a display of the I/O circuit **148** of the user device **112**). Further functionality of the user interface generation circuit **185** as well as example user interfaces will be described further herein.

The I/O circuit **180** is structured to receive communications from and provide communications to the user of the user computing device **112** (e.g., the user). In this regard, the I/O circuit **180** is structured to exchange data with the processing circuit **178** to provide output to the user and to receive input from the user. As a result, the I/O circuit **180** may include a display that may be manipulated by the application. In some embodiments, the I/O circuit **180** may also include a keyboard, a mouse, a joystick, a touch screen, touch areas, soft keys, a microphone, a speaker, a vibration mechanism, a sensor, a RFID scanner, or other input/output devices described herein.

The recipient computing device **116** is structured to interface with the provider computing system **104** and/or the user computing device **112** by way of the network to receive an HCP file, in response to the user device **112** requesting the recipient computing device **116** receive access to the HCP file. In operation, the recipient computing device **116** may receive or request access to one or more HCP files and then receive or be provided with access to the HCP files. Therefore, the recipient computing device **116** can be one or more of a mobile phone, a tablet computer, a laptop computer, a smart watch, a server computer system, and any other

15

internet-connected device. In this regard, the recipient computing device **116** may be a computing system associated with the recipient address described herein. For example, the recipient address may be an email of the recipient and the recipient computing device **116** may be multiple computing devices the recipient uses to access their email and therefore the HCP file (e.g., the recipient's smart phone and the recipient's laptop, and so on). Further, the recipient computing device **116** is shown to include a network interface circuit **186**, a processing circuit **188**, and the input/output (I/O) circuit **190**.

The network interface circuit **186** is structured to establish connections with the provider computing system **104**, the covered entity computing system **108**, the user computing device **112**, the consumer database **120**, and the HCP database **124** by way of the network **118**. The network interface circuit **186** may be similar to the network interface circuit **126** and include program logic and/or hardware-based components that connect the recipient computing device **116** to the network **118**. For example, the network interface circuit **186** may include any combination of a wireless network transceiver (e.g., a cellular modem, a broadband modem, a Bluetooth transceiver, a Wi-Fi transceiver, a Li-Fi transceiver, etc.) and/or a wired network transceiver (e.g., an Ethernet transceiver). In some embodiments, the network interface circuit **186** includes the hardware and machine-readable media structured to support communication over multiple channels of data communication (e.g., wireless, Bluetooth, near-field communication (NFC)). In some embodiments, the network interface circuit **186** includes cryptography logic and capabilities to establish a secure communications session.

The processing circuit **188**, as shown, may comprise a memory **192** and a processor **194**. The memory **192** includes one or more memory devices (e.g., RAM, NVRAM, ROM, flash memory, hard disk storage, etc.) that store data and/or computer code for facilitating the various processes described herein. That is, in operation and use, the memory **192** stores at least portions of instructions and data for execution by the processor **194** to control the processing circuit **188**. The memory **192** may be or include tangible, non-transient volatile memory and/or non-volatile memory. The processor **194** may be implemented as a general-purpose processor, an application specific integrated circuit (ASIC), one or more field programmable gate array (FPGAs), a digital signal processor (DSP), a group of processing components or other suitable electronic processing components.

The I/O circuit **180** is structured to receive communications from and provide communications to the user of the recipient computing device **116** (e.g., a recipient). In this regard, the I/O circuit **190** is structured to exchange data with the processing circuit **188** to provide output to the recipient and to receive input from the recipient. In some embodiments, the I/O circuit **180** may include a display, a keyboard, a mouse, a joystick, a touch screen, touch areas, soft keys, a microphone, a speaker, a vibration mechanism, a sensor, a RFID scanner, or other input/output mechanisms.

Still referring to FIG. 1, the consumer database **120** is a repository that receives, stores, and manages various consumer information associated with multiple consumers. To do so, the medical consumer database **120** can be structured according to various database types, such as, relational, hierarchical, network, flat, point-in time, and/or object relational. In operation, the consumer database is configured to store consumer information such as IP addresses of consumers who watched a video, IP addresses of consumers

16

who visited a specific webpage, names of consumers who live within a specific state, consumer demographic information (e.g., income information, gender information, ethnicity information), and the like. Further, the consumer information within the consumer database may be used to generate health information requests by the provider computing system **104**. In this regard, the provider computing system **104** may generate lists of consumers for which health information is sought, populate the health information request with the list, and provide the health information request to the covered entity computing system **108**, which may return deidentified health information of corresponding consumers. In some embodiments, the consumer database **120** is operated by a third-party and communicates with the provider computing system **104**, the covered entity computing system **108**, and/or the user computing device **112** over the network **118**. For example, the provider computing system **104** may send a consumer information request to the consumer database **120** including a list of consumers or HCPs for which consumer information is sought. In response, the consumer database **120** may provide consumer information associated with the HCPs to the provider computing system **104**. In other embodiments, the provider computing system **104**, the covered entity computing system **108**, or the user computing device **112** include the consumer database **120**.

Likewise, the HCP database **124** is a repository that receives, stores, and manages various HCP information associated with multiple HCPs. To do so, the HCP database **124** can be structured according to various database types, such as, relational, hierarchical, network, flat, point-in time, and/or object relational. In operation, the HCP database **124** is configured to store HCP information such as HCP address or other HCP location information (e.g., address of the HCP's employer, zip code of the HCP or the HCP's employer, etc.), HCP national provider identifier (NPI), Veeva HCP Identifier (VID), HCP specialties, HCP type (e.g., Nurse Practitioner, Doctor of Medicine, etc.), and the like. Further, the HCP database **124** may provide HCP information to the provider computing system **104** to be included in health information requests or to be matched with health information. For instance, the provider computing system **104** may receive the health information (e.g., Rx information including the name of the prescribing physician) from the covered entity computing system **108** and match it with the HCP information (e.g., the name of the HCP, the HCP NPI, the HCP specialty, etc.). The matched information may then be included in a generated HCP file. In some embodiments, the HCP database **124** is operated by a third-party and communicates with the provider computing system **104**, the covered entity computing system **108**, and/or the user computing device **112** over the network **118**. In other embodiments, the provider computing system **104**, the covered entity computing system **108**, or the user computing device **112** include the HCP database **124**.

Similarly, the one or more national health databases **125** are each a repository that receives, stores, and manages various government health information (e.g., national health insurance (e.g., Medicare, Medicaid) claims information (e.g., Mx and/or Rx claims information), universal health care (e.g., MediFund of Singapore, etc.) or coverage claims information (e.g., Mx and/or Rx), and the like. Further, the government health information may include location information (e.g., zip code, country, city, state), medical product information, national health care statistics (e.g., Rx is from Medicare part A, Medicare part B, Medicare part C, Medi-

care part D, etc.), HCP information (e.g., NPI of the HCP who prescribed a specific Mx or Rx.), deidentified PHI, and the like.

Additionally, the national health database **125** can be structured according to various database types, such as, relational, hierarchical, network, flat, point-in time, and/or object relational. In operation, the national health database **125** is configured to store, manage, and provide government health information such as national health insurance Rx claims information, national health insurance Mx claims information, and the like. In some embodiments, the one or more national health databases **125** are each operated by a computing system associated with a government healthcare agency (e.g., the Center for Medicare & Medicaid Services (CMS)). In this regard, the government health information may include a first portion associated with Rx claims information and a second portion associated with Mx claims information. In other embodiments, the national health databases **125** are each a part of the covered entity computing systems **108**. Further, the government health information may include location information (e.g., zip code, country, city, state), medical product information, national health care statistics (e.g., Rx is from Medicare part A, Medicare part B, Medicare part C, Medicare part D, etc.), HCP information (e.g., NPI of the HCP who prescribed a specific Mx or Rx.), deidentified PHI, and the like.

Referring now to FIGS. 2A-2B, a method **200** of analyzing longitudinal health information and generating a dynamically structured HCP file (also referred to as a dynamically structured electronic file) is shown, according to an example embodiment. Method **200** can be carried out by the system of FIG. 1. More particularly, the method **200** can be carried out by the processing circuit **128** of the provider computing system **104** and through communication with the covered entity computing system **108**, the user computing device **112**, and the recipient computing device **116**.

Method **200** commences at step **204** at which the provider computing system **104** receives HCP information from the HCP database **124**. The HCP information may be used to populate a health information request and may be received along with other information. For example, at step **204**, the provider computing system **104** may receive other information used to populate the health information request such as consumer information (from the consumer database **120**), timeframe information, medical product information (from the medical product database), and other information described herein as being included in the health information request. In some embodiments, the provider computing system **104** receives HCP information (or the other information with which the health information request is populated) from the HCP database **124** at specific time intervals (e.g., the same as or different than the specific intervals at which the health information requests are generated). In further embodiments, the provider computing system **104** may receive information used to populate the health information request (e.g., consumer information) at step **204** and receive the HCP information at or after another step in the method **200** (e.g., prior to or after step **228** as will be discussed further herein).

Once the provider computing system **104** has received the HCP information, the method **200** proceeds to step **208** at which the provider computing system **104** generates a health information request. As described herein, the health information request may be generated and populated to include the information received at step **204** such as the HCP information, the consumer information, the medical product

information, and the timeframe information. The health information request **208** may request health information that is associated with or matches the information included in the health information request. In some embodiments, the health information request **208** may request health information that is not associated with or does not match the information included in the health information request.

In some embodiments, at step **208**, the provider computing system **104** may generate multiple health information requests at step **208**. For example, the provider computing system **104** may generate a health information request for each covered entity computing system from which health information is sought. In other embodiments, the provider computing system **104** may generate a single, reused, health information request that is sent to each covered entity computing from which health information is sought.

Additionally, at step **208**, the provider computing system **104** may generate the health information request, at a specific time interval or rate (e.g., once a day, once a week, once a month, every 5 minutes, etc.). In some embodiments, the specific time interval or rate at which the health information request is generated is different than the specific time interval or rate at which the HCP information, medical product information, and/or consumer information is received. In other embodiments, the specific time interval or rate at which the health information request is generated is the same as the specific time interval or rate at which the HCP information, medical product information, and/or consumer information is received. By doing so, the provider computing system **104** may generate an updated health information request including the most recently received HCP, medical product, and/or consumer information.

Once the provider computing system **104** has generated the health information request, the method **200** proceeds to step **212** at which the provider computing system **104** provides the health information request to one or more covered entity computing systems (e.g., the covered entity computing system **108**). In some embodiments, the provider computing system **104** may provide the health information request over the network **118** via the network interface circuit **126** to the one or more covered entity computing systems **108**.

Once the provider computing system **104** has provided the health information request to the one or more covered entity computing systems, the method **200** proceeds to step **216** at which health information is received from the one or more covered entity computing systems **108**. The health information may be received in response to the provider computing system **104** providing the health information request to the one or more covered entity computing systems and may be or include Rx information, Mx information, medical product information, transaction information, diagnosis information, and other health information described herein. For example, the health information may be prescription claims (Rx) information that includes deidentified PHI, medical product information, diagnosis information (e.g., a diagnosis code), HCP information (e.g., an NPI of a prescribing HCP), date/time information, facility or health care system location information, and transaction information. In another example, the health information may be medical claims (Mx) information that includes deidentified PHI, medical product information, diagnosis information (e.g., a diagnosis code), HCP information (e.g., an NPI of a prescribing HCP), date/time information, facility or health care system location information, and transaction information. In another example, the health information may be national health insurance claims information or electronic health record

information including deidentified PHI, medical product information, diagnosis information (e.g., a diagnosis code), HCP information (e.g., an NPI of a prescribing HCP), date/time information, facility or health care system location information, and transaction information. Further, the received health information may correspond to or be associated with the HCP information, the consumer information, and/or the medical product information included in the health information request. For example, the health information received at step 216 may be matched with or include consumer information for each individual consumer of each prescription. For example, each piece of Rx information may be matched with demographic information associated with the consumer of the prescription. Further, the received health information may include deidentified PHI, transaction information, date/time information, and medical product information.

In some embodiments, consumer information is received from the consumer database 120 at or after step 216 and matched with the HCP information. For example, the provider computing system 104 may receive consumer information from the consumer database 120 and match the consumer information with HCPs. In this regard, the consumer information may be organized by HCP such that HCP insights or statistical values described herein may be determined from the consumer information.

As described herein, the PHI may be deidentified, by the one or more covered entity computing systems 108, in a variety of ways including through encryption, tokenization, and using a hash function. In all cases, the one or more covered entity computing systems (e.g., the covered entity computing system 108) may use a consistent form of deidentification such that the same PHI input will return the same hash value or token. For example, if a piece of health information included the name "John Smith," a resulting deidentified hash value may be "1234321." To be consistent, each of the one or more covered entity computing systems, when identifying the name "John Smith," may then return the hash value "123431." By using the same hash function across the one or more covered entity computing systems, the provider computing system 104 can make additional connections between the received health information and generate HCP insights and statistical values. Further, less processing power is required by the provider computing system 104 as most of the health information matching is completed by the one or more covered entity computing systems prior to the health information being received by the provider computing system 104.

Once the provider computing system 104 has received the health information from the one or more covered entity computing systems 108, the method 200 proceeds to step 220 at which the provider computing system 104 determines if the received health information includes duplicate health information. To determine if the health information includes duplicate health information, the provider computing system 104 may parse each piece of the health information and determine if multiple pieces have one or more matching or at least partially matching information points. For example, the provider computing system 104 may determine that two pieces of health information have the same deidentified name hash value (e.g., "1234321"), the same date/time information, and/or the same medical product information. Then, based on these three matching information points, the provider computing system 104 may determine that the two pieces of health information are duplicates. In some embodiments, the provider computing system 104 may determine that multiple pieces of health information are duplicates if

the date/time information, the deidentified PHI, and the medical product information of each piece of health information is the same.

Additionally, the provider computing system 104 may determine that multiple pieces of health information are substantially duplicates if each piece of health information includes one or more matching information points, but one of the pieces of health information includes more or different information points from the other pieces of health information. For example, the provider computing system 104 may determine that two pieces of health information have the same deidentified name hash value (e.g., "1234321"), the same date/time information, and the same medical product information, but that only one of the pieces of health information includes diagnosis information. Then, based on these three matching information points and the one additional information point (the diagnosis information), the provider computing system 104 may determine that the two pieces of health information are substantially duplicates.

If, at step 220, the provider computing system 104 determines that the received health information includes duplicate health information (or substantially duplicate health information), the method 200 proceeds to step 224 at which the provider computing system 104 modifies the received health information to remove the (substantially) duplicate health information. If the received health information includes multiple pieces of duplicate health information, the provider computing system 104 may modify the health information to remove or delete each excess piece of duplicate health information leaving only a single copy or piece of the duplicate health information. Alternatively, if the received health information includes multiple pieces of substantially duplicate health information, the provider computing system 104 may modify the health information to combine each piece of substantially duplicate health information leaving only a single, combined, piece of the substantially duplicate health information including all the information points of each piece of the substantially duplicate health information.

If, at step 220, the provider computing system 104 determines the received health information does not include (substantially) duplicate health information, or after the provider computing system modifies the health information at step 224, the method 200 proceeds to step 228. At step 228, the provider computing system 104 receives an HCP file generation request (also referred to as an electronic file generation request) from the user computing device 112. As described herein, the HCP file generation request may indicate one or more insights or statistical values (e.g., median quantity dispensed of a medical product, the number of times switched from or switched to prescribing or administering the medical product, percentage of generic medical products prescribed or administered versus name brand medical products, etc.) associated with the HCP that are to be calculated as well as include a file destination address to which the HCP file is to be provided. In some embodiments, the HCP file generation request may also identify or include a specified file format that the HCP file is to be. For example, the HCP file generation request may indicate the HCP file is to be generated as a .CSV file delimited with commas.

In some embodiments, the HCP file generation request may also identify the HCP and health information to be used in generating the HCP file. For example, the HCP file generation request may indicate a specific medical product (e.g., drug X) or medical product group for which health information is to be included and used in generating the HCP file.

21

In some embodiments, the HCP file generation request may include or indicate a specific time interval or rate at which an HCP file is to be generated. For example, the HCP file generation request may indicate that an HCP file is to be generated, based on specific health information and HCP information every month. The provider computing system 104 may then repeat the method 200 from steps 238-268 based on the HCP file generation request. In other embodiments, the provider computing system 104 may receive an HCP file generation request at the specific time interval or rate and perform steps 238-268 based on the received HCP file generation request. Additionally, the provider computing system 104 may generate a health information request (e.g., at step 208) at a first specific time interval or rate and may receive or generate an HCP file generation request at a second specific time interval or rate, that is different from the first specific time interval or rate. Because the health data may be received at a different time interval than the HCP file generation request, the provider computing system 104 may act as a source of centralized health information. For example, the provider computing system 104 may receive health information at a time interval faster than that of the HCP file generation request and continually be generating HCP files with new, updated health information.

Once the provider computing system 104 has received the HCP file generation request from the user computing device 112, the method 200 proceeds to step 232 at which the provider computing system 104 matches each piece of the HCP information with one or more pieces of the health information based on the HCP file generation request. The provider computing system 104 may match the HCP information with the health information by determining that one or more information of the HCP information match that of the health information. For example, the provider computing system 104 may determine that the health information includes a national provider identifier (NPI) of a prescribing physician. The provider computing system 104 may then match the NPI of the prescribing physician with the NPI of the HCP information. In this regard, the health information may be sorted based on each HCP. In some embodiments, the NPI may be a hashed or tokenized NPI such that the health information may be sorted based on each NPI of each HCP. In other embodiments, the provider computing system 104 may match one or more of a HCP name, an HCP address, an HCP specialty, or an Rx number of the health information with that of the HCP information.

In some embodiments, step 232 and the provider computing system 104 matching the HCP information with the health information may happen prior to receiving the HCP file generation request. For example, the provider computing system 104 may match the health information with the HCP information directly after step 220 and/or step 224 at which the duplicate information is removed from the health information. In another example, the provider computing system 104 may match the health information with the HCP information directly after step 216 in which the health information is received from the covered entity computing system 108 and the duplicate information may then be removed from the matched health information.

Once the provider computing system 104 has matched the HCP information with the health information, the method 200 proceeds to step 234 at which the provider computing system 104 filters at least one of the matched health information or the matched HCP information based on the HCP file generation request. For example and as discussed herein, the HCP file generation request may indicate a specific medical product group (e.g., drugs X, Y, and Z) for which

22

health information is to be included and used in generating the HCP file. As a result and at step 234, the provider computing system 104 may filter, process, or refine the health information to remove the health information including medical product information that is not within the specific medical product group (e.g., drugs that are not X, Y, or Z or drugs that are not Brand X, Brand Y, or Brand Z) and resulting in health information that only includes medical product information that is within the specific medical product group. In another example, the HCP file generation request may indicate a specific specialty of HCP (e.g., cardiology) for which HCP information is to be included in generating the HCP file. Then, at step 232, the provider computing system 104 may filter, process, or refine the HCP information to remove the HCP information including specialty information that are not cardiology and resulting in HCP information that only includes specialty information that is cardiology. If matching health information is removed during the filtering process, the matched HCP information may be removed as well, assuming there is no remaining health information matched with the HCP information. In contrast, if matching health information is removed during the filtering process and there is remaining health information matched with the HCP information, the HCP information may not be removed. Further, if matching HCP information is removed during the filtering process, the matched health information may be removed as well, assuming there is no remaining HCP information matched with the health information.

In some embodiments, the order of steps 232 and 234 is reversed such that the provider computing system 104 filters and then matches the HCP information and the health information. In other embodiments, steps 232 and 234 may occur at the same time such that the provider computing system 104 filters and matches the HCP information and the health information at the same time (i.e., filtering may be done by matching the medical product information of the health information with that of the medical product group and likewise for diagnosis information and diagnoses groups). In this regard, it should be understood that the steps 232 and 234 are interchangeable and may occur in any order or at the same time.

Once the provider computing system 104 has filtered the matched HCP information and health information, the method 200 proceeds to step 236 at which the provider computing system 104 determines one or more HCP statistical values (e.g., insights). The HCP statistical values may be calculated or determined using the matched (and/or filtered) health information (e.g., Rx or Mx information), the matched (and/or filtered) HCP information, and, in some embodiments, the associated consumer information from step 204. Possible HCP statistical values will be discussed further herein but may include: a number of total prescriptions for a medical product for each HCP, the number of new prescriptions (i.e., the first time a patient has received a prescription of a specific medical product) for each HCP, the number of new-to-brand prescriptions (i.e., the first time a patient has received a prescription of a specific brand of a medical product) for each HCP, the number of total diagnoses of a medical condition for each HCP, the number of total patients each HCP has seen, the total number of medical claims processing for each HCP, the total number of prescriptions filled and medical claims processed for each HCP, the number of new to brand medical claims, the number of new to brand medical and prescription claims, the number of total patients on Medicare part B, part A, part C, and other national health care demographics (e.g., number of

23

patients on Medicare, number of patients on Medicaid, etc.), patient demographics (e.g., % per specific age range, % per gender, % per income range, % per payment type) for each HCP, median quantity of medical product dispensed for each HCP, median days supply of each medical product dispensed for each HCP, number of times switching from a medical product for each HCP, number of times switching to a medical product for each HCP, and average time between switching to/from a medical product for each HCP.

For example, the HCP file generation request may indicate that the HCP file is to include patient demographics for each individual HCP, the median quantity of drug X that is dispensed by each HCP, and the number of times each HCP has switched from the drug X to drug Y. As a result and at step 236, the provider computing system 104 may determine patient demographics for each individual HCP based on the matched HCP information, the matched health information, and/or the received consumer information. For instance, the provider computing system 104 may parse the matched health information for each HCP and sum up the payment types (e.g., credit card, check, cash, Medicare payment, Medicaid payment, Commercial insurance payment, etc.) within the transaction information and then divide the summed number of each payment type by the total number of payments resulting in a % per payment type. The provider computing system 104 may further parse the matched health information for Rx information for drug X and determine the quantity of drug X dispensed in each prescription. Then, the provider computing system 104 may calculate the median quantity based on the quantity of drug X in each transaction. Lastly, the provider computing system 104 may parse the matched health information and search for a switch from drug X to drug Y. To determine a switch, the provider computing system 104 may determine a patient for an HCP was prescribed drug X by the HCP at a first point in time and then was prescribed drug Y by the HCP at a second point in time, after the first point in time. In some embodiments, the switch is only determined/counted if the patient is not prescribed drug X at or after the second point in time. The provider computing system 104 may then determine the total number of switches from drug X to drug Y for each HCP.

In another example, the HCP generation request may indicate that the HCP file is to include an adoption profile for the HCP for a specific medical product or group of medical products. The adoption profile may indicate the number of times the HCP switched to or from prescribing or administering the specific medical product and the drugs that the HCP switched to or from. Switching from a medical product indicates that the HCP previously prescribed a specific drug (e.g., drug X) for a given condition but has now begun prescribing another drug (e.g., drug Z), for the same condition. For example, the adoption profile for the HCP may indicate that 30% of the time the HCP switched from prescribing or administering drug X. Switching to a medical product indicates that the HCP previously prescribed a specific drug (e.g., drug J) for a given condition but has now begun prescribing another drug (e.g., drug K) for the same condition. For example, the adoption profile for the HCP may indicate that 30% of the time the HCP switched to prescribing or administering drug K. In other embodiments, the adoption profile may indicate a number of switches for each drug as compared to a percentage. To determine, the adoption profile for a specific medical product and HCP, the provider computing system 104 may determine each drug from which the HCP switched to or from the specific drug as well as the number of switches for each drug. Then, the provider computing system 104 may add up the total number

24

of switches to or from the specific medical product, and divide each value by the total to get a percentage.

Once the provider computing system 104 has determined the one or more HCP statistical values or insights, the method 200 proceeds to step 240 at which the provider computing system 104 generates an HCP file including (e.g., populated with) at least a portion of the matched HCP information, at least a portion of the matched health information, and/or the HCP statistical values. The HCP file may be populated with at least a portion of the matched HCP information and the matched health information in that the HCP file may include some of the matched HCP information and the matched health information and omit other portions of the HCP information and the matched health information. For example, the HCP file may specifically omit any deidentified PHI of the matched health information. In some embodiments, the HCP file is populated with at least a portion of the matched HCP information and the HCP statistical values, but not the matched health information (as it is represented by the HCP statistical values).

Additionally and as described herein, the provider computing system 104 may structure the HCP file based on the HCP file generation request, the destination address of the HCP file generation request, and the specific file type of the HCP file generation request. For example, the provider computing system 104 may populate the HCP file with the generated HCP statistical values and the matched HCP information in a specific order based on the HCP file generation request and the recipient address of the HCP file generation request. For instance, if the HCP file is to be provided to the user computing device 112 (e.g., the destination address is the IP address of the user computing device 112), the provider computing system 104 may structure the HCP file to include a specific header for each HCP statistical value and for each piece of matched HCP information as well to include a single row for each HCP. In comparison, if the HCP file is to be provided to another portion of the provider computing system 104 (e.g., Veeva Nitro), the provider computing system 104 may structure the HCP file to include a specific row for each HCP statistical value and multiple rows for each HCP.

The HCP file may be generated as a specific file type as decided by the user in the HCP file generation request. For example, the HCP file may be generated as a CSV file, an Excel File (XLSX file), an XML file, a JSON file, and the like. By allowing the user to decide the specific file type of the HCP file in the HCP file generation request and not only generating the HCP file as one specific file type, the provider computing system 104 may provide for improved shareability of the HCP file as well as use less processing power and memory. For example, in situations where the HCP file is only generated as a specific, non-dynamic, file type, third-party computing systems and recipient computing devices (e.g., the recipient computing device 116) may have to perform additional file modification and conversion or request the provider computing system 104 to do so, which requires additional processing power, memory (to keep two files), and time. In comparison, the system 100 and the provider computing system 104 allow the user to specify the file type of the HCP file through the HCP file generation request. Accordingly, the user can specify the wanted file type the first time and the provider computing system 104 will generate the HCP as the specified file type. As a result, no additional file modification or conversion is required reducing processing power usage, memory usage, and overall processing time for the provider computing system 104.

25

Once the provider computing system **104** has generated the HCP file, the method **200** proceeds to step **242** at which the HCP file is stored in the account database **132**. The provider computing system **104** may store the HCP file in the account database **132** in association with the user's account. In some embodiments, the provider computing system **104** may further retrieve the HCP file from the account database **132** and provide the HCP file to the user computing device **112** in response to the user requesting access to the HCP file. In some embodiments, the user and the user computing device **112** must first provide login credentials (e.g., a username and a password) and be authenticated before the provider computing system **104** will provide the HCP file to the user computing device **112**.

Once the provider computing system **104** has stored the HCP file in the account database **132**, the method **200** proceeds to step **244** at which the HCP file is provided to the file destination address of the HCP file generation request. The file destination address may be any type of address or destination (e.g., IP address, web address, email address, file transfer protocol (FTP) address, and the like) to which the HCP file may be electronically transmitted or provided. In one example, the file destination address may be another server or circuit of the provider computing system **104** that is configured to receive the HCP file (e.g., Veeva Nitro®). In another example, the file destination address may be a web address of the Amazon Simple Storage Service® cloud storage web service, a web address of Snowflake, a web address of Microsoft Azure, etc. In another example, the file destination address may be the IP address of the user computing device **112** and the HCP file may be provided to the user computing device **112**.

Once the provider computing system **104** has provided the HCP file to the file destination address, the method **200** proceeds to step **246** at which the provider computing system **104** receives a file share request from the user computing device **112**. The file share request may identify the HCP file (e.g., of a multiple of HCP files) and indicate that the user (of the user computing device **112**) would like to share the HCP file with one or more recipients (e.g., of the recipient computing device **116**) up until a specified point in time. As a result, the file share request may include a recipient address and a file share expiration date and/or time. The recipient address may be any address or destination (e.g., IP address, web address, email address, file transfer protocol (FTP) address, and the like) to which the HCP file may be electronically transmitted, provided, and/or accessed by up until a specific point in time (e.g., at or after the file share expiration date and/or time). In one example, the recipient address may be the IP address of the recipient computing device **116** and the HCP file may be transmitted to the recipient of the recipient computing device **116**. In another example, the recipient address may be the email address of the recipient of the recipient computing device **116**.

The file share expiration date and/or time is a specific date and/or time or time interval that defines a specific date and time at which the HCP file is no longer to be shared with the recipient address. For example, the file share expiration date and/or time may indicate the HCP file is no longer to be shared with the recipient address on Dec. 7, 2021 at 2:00 PM. Then, prior to Dec. 7, 2021 at 2:00 PM, the provider computing system **104** may share (e.g., provide, electronically transmit in response to a request, provide access via a web server, etc.) the HCP file with the recipient address. However, after Dec. 7, 2021 at 2:00 PM, the provider computing system **104** may not share (e.g., provide, elec-

26

tronically transmit in response to a request, provide access via a web server, etc.) or even deny access (e.g., display a message indicating as such and/or telling the recipient to contact the user of the user computing device) to the HCP file by the recipient address. In another example, the file share expiration date and/or time may indicate the HCP file is no longer to be shared with the recipient address on Dec. 7, 2021 and not specify a time. As a result, the recipient address may not have access to the HCP file on Dec. 7, 2021 at 12:00 AM. In some embodiments, the provider computing system **104** may specify the expiration date and/or time as being in the same time zone as the user (e.g., the user computing device **112**).

Once the provider computing system **104** has received the file share request from the user computing device **112**, the method **200** proceeds to step **256** at which the provider computing system **104** provides an indication to a recipient computing device (e.g., the recipient computing device **112**) associated with the recipient address. The indication may identify the HCP file (e.g., the name of the HCP file, the location of the HCP file, and the like) as well as the file share expiration date/time. The indication may be used to simply provide the recipient notice that they now have access to the HCP file. In some embodiments, the indication may further include a Uniform Resource Locator (URL) or web address that the recipient may select (e.g., via a user interface on the recipient computing device **116**) to be navigated to a web application in which the recipient may have access (i.e., download, view, access) the HCP file and which may be provided to the recipient computing device **116** in response to authenticating the recipient address as well as confirming the file share expiration date and/or time is not met/passed. In one example, the indication may be an email that is provided to a recipient email address and includes an HCP file URL. The HCP file URL may further include an expiration parameter (e.g., signature) that is associated with the file share expiration date and/or time. For example, the expiration parameter may be included in the URL and deactivate the URL after the file share expiration date and/or time. As a result, if the recipient selects the URL including the expiration parameter after the file share expiration date and/or time, the URL may be deactivated and not route the recipient to the web application or the HCP file.

Once the provider computing system **104** provides the indication to the recipient computing device **116**, the method **200** proceeds to step **258** at which the provider computing system receives a request to access the HCP file from the recipient computing device **116**. The request to access the HCP file may identify the HCP file as well as the recipient address through which the recipient computing device **116** is requesting access. For example, the request may be received and generated in response to the recipient of the recipient computing device selecting the HCP file URL of the indication. In some embodiments, the request to access the HCP file is received after authenticating the recipient address of the recipient computing device **116** (e.g., after determining that recipient address is authorized to access the HCP file).

Once the provider computing system **104** receives the request to access the HCP file, the method **200** proceeds to step **260** at which the provider computing system **104** determines if the present date/time is past the file share expiration date/time. If, at step **260**, the provider computing system **104** determines that the present date/time is not past the file share expiration date/time, the method **200** proceeds to step **264** at which the provider computing system **104** allows access to the HCP file by the recipient computing device **116**. As described herein, the provider computing

system 104 may allow access to the HCP file by the recipient computing device 116 in multiple ways. For example, the provider computing system 104 may provide or electronically transmit the HCP file to the recipient computing device 116. In another example, the provider computing system 104 may provide a web application and a web interface to the recipient computing device 116 through which the recipient may view and/or download the HCP file. In comparison, if at step 260, the provider computing system 104 determines that the present date/time is past the file share expiration date/time, the method 200 proceeds to step 268 at which the provider computing system 104 denies access to the HCP file by the recipient computing device 116. The provider computing system 104 may allow deny to the HCP file by the recipient computing device 116 in multiple ways. For example, the provider computing system 104 may provide or electronically transmit a notification to the recipient computing device 116 indicating that the recipient is not authorized to access the HCP file. In another example, the provider computing system 104 may further notify the user of the user computing device 112 that the recipient is attempting to access the HCP file and inquire if the file share expiration date/time should be extended.

Referring now to FIGS. 3-7, user interfaces shown to the user of the user computing device 112 during the method 200 and/or the method 900 are shown, according to example embodiments. As described herein, the user interfaces of FIGS. 3-7 may be one or more of web interfaces generated by the provider computing system 104 and rendered by the user computing device 112 as part of a web application or graphical user interfaces downloaded and generated by the user computing device 112 as part of a software application (e.g., mobile application, etc.). Further, the user interfaces shown on FIGS. 3-7 allow for communication between the user and the provider computing system 104 via the user computing device 112 (specifically via the I/O circuit 180). Through interaction with the various user interfaces, the user may provide user input, feedback, and other information requested by the provider computing system 104.

Referring now to FIG. 3, an information selection page 300, which can be displayed on a display of the I/O circuit 180 of the user computing device 112, is shown. In general, the information selection page 300 provides the user with multiple information types that can be analyzed and used to generate an HCP file generation request and therefore an HCP file. For example, through the information selection page, the user can decide between analyzing patient specific information, HCP specific information, prescription sales information, and national prescription information.

As shown, the information selection page 300 is shown to include a pull data button 304, a job manager button 308, a partner manager button 312, and a setup button 316. The pull data button 304 is a selectable (e.g., clickable, pressable) button that, when selected, generates and navigates the user to the information selection page 300 to select the information they would like to use and the type of file they would like to generate. In some embodiments, the data pull button 304, when selected, navigates the user to a web page that displays the information selection page 300. The job manager button 308 and the partner manager button 312 are each also selectable buttons that, when selected, navigate the user to other pages that will be described further herein. Similarly, the setup button 316 is a selectable button that, when pressed by the user, navigates the user to a setup/preferences page (not shown). The setup/preferences page provides the user with information associated with account preferences such as login information and medical device information.

Still referring to FIG. 3, the information selection page 300 is shown to include an information selection menu or checkbox 318 with multiple information options or checkboxes 320-332 and a next button 336. Through interaction with the information selection menu 318 and the information options 320-332, the user of the user computing device 112 can indicate the type of information that will be analyzed and used in the HCP file generation process (e.g., the method 200). As shown, the information selection menu 318 includes multiple selectable options including a patient information option 320, an HCP information option 324, a prescription sales information option 328, and a national prescription information option 332. Only the HCP information option 324 will be described in detail herein. The HCP information option 324 is a selectable or clickable option of the information selection menu 318 through which the user of the user computing device 112 can indicate they would like to generate an HCP file and an HCP file generation request (as discussed in regard to the method 200). For example, the user of the user computing device 112 may select the HCP information option 324 and then select the selectable next button 336. In response, the user computing device 112 may send an indication to the provider computing system 104 that the user is about to generate an HCP file generation request. In response to the indication, the provider computing system 104 may provide medical device information and other information to the user computing device 112 that will be used in generating the HCP file generation request and as will be discussed further herein.

Referring now to FIGS. 4A-4C, an HCP file generation request page 400 (or HCP request page 400, for short) is shown. The HCP request page 400 can be displayed on a display of the I/O circuit 180 of the user computing device 112 and, in general, provides the user with an interface (e.g., a page including options, information, and other features that will be described herein) to generate an HCP file generation request. For example, through interaction with the HCP request page 400 the user of the user computing device 112 can generate an HCP request indicating how the health information, HCP information, and consumer information should be filtered; identifying the file destination, identifying the file type of the HCP file, and indicating the one or more HCP statistical values or insights that are to be generated. To access or be navigated to the HCP request page 400, the user of the user computing device 112 may navigate to the information selection page 300 (e.g., by selecting the pull data button 304), then select the HCP information option 324 of the information selection menu 318, and finally select the next button 336.

As shown, the HCP request page 400 includes the pull data button 304, the job manager button 308, the partner manager button 312, and the setup button 316, which all provide for navigation between various pages as described herein. The HCP request page 400 further includes multiple save buttons 404 (e.g., "Save" and "Save As"), a submit button 408, and a health and HCP information filter menu 410.

The save buttons 404 allow the user of the user computing device to save the current HCP file generation request such that they can return to it at another point in time. The "Save As" button may provide a popup menu or box through which the user can name the current HCP file generation request or overwrite another saved HCP file generation request. The "Save" button may simply save a pre-named or selected HCP file generation request such that the user can return to it under that pre-selected name at another point in time. When the user selects either of the save buttons 404, the user

29

computing device **112** may generate an HCP file generation request, with the currently selected information and details of the HCP request page **400** (including empty fields where the user has not made a selection yet) and provide the HCP file generation request to the provider computing system **104** with an indication that it is to be saved (but not used to generate an HCP file). The provider computing system **104** may receive the indication and the HCP file generation request and save the HCP file generation request (and accompanying selected information and name) in association with the account of the user. For example, the provider computing device may provide the HCP file generation request and accompanying information to the account database **132** for storage. The user of the user computing device **112** may then return to the saved HCP file generation request through selection of the job manager button **308** as will be described further herein.

The health and HCP information menu **410** is a menu that includes a diagnosing option **412**, a treating option **416**, and a diagnosing and treating option **420** and through which the user of the user computing device **112** can indicate one or more filters of the matched health information and HCP information. For example, through interaction with the health and HCP information menu **410**, the user can generate an HCP file generation request that filters or instructs the provider computing system **104** to filter the matched health information and HCP information for HCPs with matched health information that includes specific diagnosing information (e.g., through selection of the diagnosing option **412**), with matched health information that includes specific treating (medical product) information (e.g., through selection of the treating option **416**), or with matched health information that includes specific diagnosing and specific treating information (e.g., through selection of the diagnosing and treating option **420**).

When the diagnosing option **412**, the treating option **416**, or the diagnosing and treating option **420** are selected, new menus or sections of the HCP request page **400** may be generated in response. For example and referring specifically to FIG. 4B, when the user selects any of the options **412-420**, the filter menu **424** may be generated and displayed, the HCP insights menu **448** may be generated and displayed, and the delivery menu **472** may be generated and displayed to the user of the user computing device **112**. Moreover, each of the menus **424-472** may be generated and include specific options, selections, or pieces of information based specifically on the selected option of the health and HCP information menu **410**. For example, in FIG. 4B the diagnosing and treating option **420** has been selected and therefore the filter menu **424** includes multiple diagnosing information filters **428** and multiple treatment or medical product filters **432**.

In comparison, if the diagnosing option **412** was selected, the filter menu **424** may include the multiple diagnosing information filters **428** but not the medical product filters **432**. Likewise, if the treating option **416** was selected, the filter menu **424** may include the medical product filters **432** but not the diagnosing information filters **428**. Additionally, once the option of the health and HCP information menu **410** has been selected, the option (in FIG. 4B the diagnosing and treating option **420**) may be emphasized or highlighted such that the option has a different color than the others. By highlighting the selected option, the HCP request page **400** may quickly and easily indicate the option that has been selected and provide such an indication to the user of the user computing device **112**.

30

The filter menu **424** is a section of the HCP request page **400** through which the user of the user computing device **112** can customize and decide the specific filters that should be applied to the (matched) HCP information and health information to be used in generating the HCP file. As shown, the filter menu **424** includes diagnosing information filters **428**, multiple new filter buttons **430**, the medical product filters **432**, the specialty information drop down box **436**, the non-prescriber checkbox or option **440**, and the pediatrician checkbox or option **444**. The diagnosing information filters **428** may be selectable options that include dual functionality. If selected once, the diagnosing information filters **428** may indicate a diagnoses group filter that will be applied to the matched HCP and health information and specifically to the diagnosis information of the matched HCP and health information. For example, in FIG. 4B the “Rheumatoid Arthritis M05+M06” may specify or indicate that the matched HCP and health information is to be filtered to only include health information that includes diagnosis information for Rheumatoid Arthritis (RA) having a diagnosis code of M05 or M06. In this way, during the method **200**, the provider computing system **104** that receives an HCP file generation request with the “Rheumatoid Arthritis M05+M06” diagnosis filter **428** may remove all HCPs and/or matched health information that does not include diagnosis information having a RA diagnosis code of M05 or M06. Thus, the resulting information that will be used to generate the HCP file will only be HCPs who have diagnosed RA with a diagnosis code of M05 or M06 and will only be for the patients who have been diagnosed with RA and a diagnosis code of M05 or M06.

Similarly, the medical product filters **432** may also be selectable options that include dual functionality. If selected once, the medical product filters **432** may indicate a medical product group filter that will be applied to the matched HCP and health information and specifically to the medical product information of the matched HCP and health information. For example, in FIG. 4B the “All Rheumatoid Arthritis” medical product information filter **432** may specify or indicate that the matched HCP and health information is to be filtered to only include HCPs matched with health information that includes medical product information for a group of medical products or drugs used to treat RA. The RA drugs or medical products may be received from the medical domain database **134** or from other databases, dictionaries, or the like that identify medical products used to treat RA. In this way, during the method **200**, the provider computing system **104** that receives an HCP file generation request with the “All Rheumatoid Arthritis” medical product information filter **432** may remove all HCPs and/or matched health information that does not include medical product information that match a product in the list of RA drugs. Thus, the resulting information that will be used to generate the HCP file will only be HCPs who have prescribed a medical product treating RA and will only be for the health information of patients who have been prescribed the medical product treating RA.

If the user double selects (e.g., double clicks) the diagnosing information filter **428** or the medical product information filter **432**, the user may be navigated to a page where they can define the diagnoses group or medical product group and therefore the specific filter. For example, if the user were to double click the “All Rheumatoid Arthritis” medical product information filter **432** the user may be navigated to a page where the user can define the specific list of drugs or medical products that define the “All Rheumatoid Arthritis” medical product group. In this way, the filters

are customizable and may be decided specifically by the user. Likewise, if the user selects the new filter button **430**, the user may be taken or navigate to a page where the user can create and define a new diagnoses group or medical product group and therefore the diagnosing information filter **428** or medical product information filter **432**, dependent on which new filter button **430** the user selected (e.g., under the “Diagnosing” header or under the “Treating” header). Further details on creating, defining, or customizing a specific filter will be described with reference to FIG. 5.

The specialty information drop down box **436** is a drop down box that is selectable to drop down a plurality of specialty filters (not shown). Once selected and dropped down, the plurality of specialty filters are selectable to indicate and choose a filter that will be applied to the (matched) HCP information and health information. For example, the user of the user computing device **112** may select the specialty information drop down box **436**, which may drop down the specialty filters “Cardiology,” “Rheumatology,” and “Neurology.” The user of the user computing device may then select “Cardiology,” which when provided to the provider computing system **104** in an HCP file generation request, may indicate the provider computing system **104** is to filter or remove all HCP information and matched health information that does not include “Cardiology” as specialty information (e.g., as a specialty). In this way, the provider computing system **104** may remove all HCPs (and corresponding HCP and health information) that is not a Cardiologist and return all Cardiologists in the HCP file. In some embodiments, the multiple specialty filters of the specialty drop down box **436** are selectable such that the user can select multiple specialties of HCPs at one time.

Similarly, the non-prescriber checkbox or option **440** and the pediatrician checkbox or option **444** are each selectable options or checkboxes that, when selected, indicate the (matched) HCP information and health information are to be filtered. For example, the non-prescriber option **440**, when not selected, may indicate the provider computing system **104** is to filter out or remove HCP information and matched healthcare information for HCPs who are not prescribers (e.g., medical interns). The provider computing system **104** may do so based on the type of the HCP included in the HCP information or based on the matched health information not including any Rx or Mx information. Likewise, the pediatrician option **444**, when not selected, may indicate the provider computing system **104** is to filter out or remove HCP information and matched healthcare information for HCPs who are Pediatricians. The provider computing system **104** may do so based on the type of the HCP or the specialty of the HCP included in the HCP information.

Still referring to FIG. 4B, the HCP insights menu **448** is shown to include HCP options or checkboxes **450**, a diagnosis statistical value or insight option or checkbox **452**, multiple treatment or medical product statistical value or insight options or checkboxes **456**, multiple patient insights or statistical value options or checkboxes **460**, a time unit selector **464**, and a timeframe drop down box **468**. Through interaction with the HCP insights menu **448**, the user of the user computing device **112** can specify the HCP information, the timeframe and time interval, and the HCP statistical values to be included in the HCP file. While all the HCP statistical values discussed with regard to step **236** of the method **200** are not shown to be included in the HCP insights menu **448**, it should be understood that the HCP insights menu **448** may include selectable options or checkboxes for

each (e.g., median quantity of each medical product specified in the medical product information filter **432** dispensed, etc.).

The HCP options or checkboxes **450** are selectable checkboxes that may be selected or unselected to specify the HCP information (e.g., NPI, VID, Specialty, HCP type, State, and/or zip) to be included in the HCP file. By selecting or deselecting each of the HCP checkboxes **450**, the user of the user computing device **112** may specify what HCP information is to be included in the HCP file.

The diagnosis statistical value checkbox **452** may include an HCP statistical value associated with the diagnosis (as defined by the diagnosis information filters **428**) that can be generated and included in the HCP file, in response to the checkbox **452** being selected. As shown, the diagnosis statistical value checkbox **452** is shown to be the total number of diagnoses for each specific diagnosis. The total number of diagnoses may be determined by adding up each piece of diagnosis information for a specific diagnosis code included in the health information matched with a single HCP. This process may be repeated for each HCP and for each diagnosis code included in the diagnosis filter **428**. In another example, the HCP insights menu **448** may include a diagnosis statistical value checkbox **452** for a total number of diagnosis overall. The total number of diagnosis may be determined by adding up diagnosis, regardless of diagnosis code, included in the health information matched with a single HCP. This process may be repeated for each HCP. In a third example, the HCP insights menu **448** may include a diagnosis statistical value checkbox **452** for a diagnosis code percentage. The diagnosis code percentage may be determined by determining the total diagnoses for each specific diagnosis and dividing it by the total number of diagnoses overall for each HCP. This process may be repeated for each specific diagnosis and for each HCP.

Likewise, the treatment or medical product statistical value checkboxes **456** may each include an HCP statistical values associated with each medical product (as defined by the medical product filters **432**) that can be generated and included in the HCP file, in response to each checkbox **456** being selected. As shown, the medical product statistical value checkbox **456** includes a number of total prescriptions for each medical product checkbox, a number of new prescriptions for each medical product checkbox, and a number of new-to-brand prescriptions for each medical product checkbox. The number of total prescriptions (TRx) for each medical product may be determined by adding up or summing the volume of prescriptions filled for patients treated by each HCP. The number of new prescriptions (NRx) for each medical product may be determined by adding up or summing each first prescription of a medical product dispensed, not a refill of a medical product, for patients treated by each HCP. The number of new-to-brand prescriptions (NBRx) may be determined by adding up or summing each of the first prescription of a specific brand of the medical product for patients treated by each HCP. Further, while not shown, the total number of medical claims processed (TMx) may be determined by adding up or summing the volume of medical claims processed for each medical product for patients treated by each HCP. While also not shown, the total number of prescriptions filled and medical claims processed (TMRx) may be determined by adding up or summing the volume of prescriptions filled and medical claims processed for each medical product for patients treated by each HCP. Likewise, the total number of new-to-brand medical claims (NBMx) may be determined by adding up or summing the first medical claim processed

of a specific brand of the medical product for patients treated by each HCP. Lastly and while not shown, the total number of new-brand-starts (NBS) may be determined by adding up or summing each of the first prescription or the first medical claim processed of a specific brand of the medical product to patients treated by each HCP. It should be understood that HCP statistical values that include Rx (e.g., NBRx, TMRx, TRx) as well as the NBS may be determined based on prescription claims (Rx) health information, and HCP values that include Mx (e.g., NBMx, TMRx, TMx) as well as the NBS may be determined based on medical claims (Mx) health information. All these calculations may be repeated for each medical product and for each HCP and may be reported as an individual number or as a percentage of a whole.

The patient statistical value checkboxes **460** may each include an HCP statistical value for patients associated with the HCP (also referred to as associated patients) (e.g., patients that have been prescribed a medical product by the HCP or been diagnosed by the HCP) that can be generated and included in the electronic file, in response to each checkbox **460** being selected. As shown, the patient statistical value checkboxes **460** includes a number of total patients checkbox, a patient age group checkbox, a patient gender checkbox, a patient income checkbox, a patient ethnicity checkbox, and a payment type checkbox. The total number of patients for each HCP may be determined by summing each individual piece of deidentified identity information (e.g., name, social security number). The % per age group of patients for each HCP may be determined by determining the age for each associated patient of the HCP based on the health information or the consumer information associated with the patient, summing up the number of patients within each age group (i.e., 0-18, 19-25, 26-40, etc.) and then dividing the number of patients within each age group by the total number of patients for each HCP. The patient gender may be determined by determining the gender for each associated patient of the HCP based on the health information or the consumer information associated with the patient, summing up the number of patients of each gender, and then dividing the number of patients of each sex by the total number of patients for each HCP. The patient income may be determined by determining the income for each associated patient of the HCP based on the consumer information associated with the patient, summing up the number of patients within each income group (i.e., \$0-10,000; \$10,001-30,000; etc.), and then dividing the number of patients of each income group by the total number of associated patients for each HCP. Patient ethnicity may be determined by determining the ethnicity for each associated patient of the HCP based on the consumer information or the health information, summing up the number of patients of each ethnicity, and then dividing the number of patients of each ethnicity by the total number of patients for each HCP. Lastly, the payment type may be determined by determining the payment type for each transaction within the health information, summing up the number of each payment type, and then dividing the number of payments of each payment type by the total number of transactions. In some embodiments gender, income, age group, ethnicity, and payment type are reported as an overall number (e.g., a sum for each) or as percentage for each.

The time unit selector **464** and the timeframe drop down box **468** are two menu options through which the user of the user computing device **112** can set the timeframe for the HCP and health information used to generate and included in the HCP file. For example, the time unit selector **464** is a

selectable option that applies the unit to the timeframe drop down box **468**. As shown, the time unit selector **464** is movable between monthly and yearly. Therefore, through interaction with the time unit selector **464**, the user can decide if they want the time frame to be measured in months or in years. Then, once the time unit selector **464** is set on the correct unit, the user can select the timeframe drop down box **468** and select the timeframe within which they want the HCP information and health information to be. For example, with the time unit selector **464** set to yearly, the user may select the time frame drop down box **468** and select the drop down option "Last 1 Year." Once set and the HCP file generation request is sent to the provider computing system **104**, the provider computing system **104** may then only use health information or HCP from within the last year. In some embodiments, the time frame drop down box **468** can include the drop down options of last 1, 2, 3 . . . 12 months, when the time unit selector **464** is on monthly and the options of last 1, 2, 3, 4, 5, 20 years, when the time unit selector **464** is on yearly.

Still referring to FIG. 4B, the delivery menu **472** is shown to include multiple file type options **476** and multiple file destination options **478**. Through interaction with the delivery menu **472**, the user of the user computing device may decide the specified file type of the HCP file as well as the file destination address. In this regard, the file type options **476** are selectable options through which the user can decide the specified file type into which the HCP file will be generated. For example, the delivery menu **472** is shown to include a XLS file type option **476** and a CSV file type option **476**. If the user selects the XLS file type option **476** and proceeds to generate submit the HCP file generation request to the provider computing system **104**, the provider computing system **104** may generate the HCP file as the specified XLS file type. Further, in some embodiments, the file type options **476**, when selected, may further indicate file destination address is the IP address of the user computing device **112**, such that the HCP file may be provided to the user computing device **112** via the internet. In other embodiments, the file destination may be indicated as being the IP address of the user computing device **112** if no file destination option **478** is selected.

The file destination options **478** are selectable options that allow the user to decide and indicate the file destination address to which the HCP file is to be provided. For example, the delivery menu **472** is shown to include a Secure FTP (SFTP) file destination option **478**, an Amazon S3® file destination option **478**, and a Veeva Nitro® file destination option **478**. If the user selects the SFTP file destination option **478**, a popup may be generated on the user computing device **112** requesting an FTP address to which the HCP file is to be provided. Once the user provides an address and submits the HCP file generation request, the provider computing device **112** may generate the HCP file, as the specified file type indicated by the selected file destination option **476** and provide the HCP file to the FTP address received from the user.

Referring to FIG. 4C, once the user has finished filling out specifying each option and field of the HCP file generation request via the HCP request page **400**, the user may select the submit button **408**. When the submit button **408** is selected, the user computing device **112** may generate and display a request popup box **480** to the user. The request popup box **480** may include a name text field **484**, a second submit button **488**, and a cancel button **490**. The user may then type or enter (e.g., via a keyboard of the input device of the I/O circuit **180**) a name of the HCP file generation

request into the name text field 484 and select the second submit button 488. Once selected, the user computing device 112 may generate an HCP file request including all of the options, details, and selections of the user on the HCP request page 400 and provide the HCP file request to the user provider computing device 112. In this regard, it should be understood that an HCP file generation request may include any of the options, features, menus, or information selections shown and described with regard to the HCP request page 400. If the user instead selects the cancel button 490, the request popup box 480 may be closed or removed from the HCP request page 400.

Referring now to FIG. 5, a medical product group selection page 500 is shown. The medical product group selection page 500 can be displayed on a display of the I/O circuit 180 of the user computing device 112 and, in general, provides the user with an interface (e.g., a page including options, information, and other features that will be described herein) to define a medical product group. For example, through interaction with the medical product group selection page 500 the user of the user computing device 112 can generate or define a medical product group which can then be displayed on the HCP request page 400 as one of the medical product filters 432 and used to filter the matched HCP information and health information. To access or be navigated to the medical product group selection page 500, the user of the user computing device 112 may double select or click one of the medical product filters 432 or select the new filter button 430 under the treatment header. If the user double selects one of the medical product filters 432, the user may be navigated to the medical product group selection page 500 which may be automatically populated with information pertaining to the selected medical product group. By contrast, if the user selects the new filter button, the user may be navigated to the medical product group selection page 500 which may not be populated with any information pertaining to a medical product group.

As shown, the medical product group selection page 500 includes the multiple save buttons 404, a medical product search bar 502, a medical product search results section 504, multiple search result filter options 516, a medical product group summary section 524, a save new medical group button 534 button, a done/submit button 536, and a cancel button 540. When the user is directed to the medical product group selection page 500, the user is typically trying to edit or create a new medical product group with one or more medical products. As a result, the medical product search bar 502 is a selectable bar with a text entry field and a search or go button. In use, the user of the user computing device 112 may type or enter a medical product identifier (e.g., name, national drug code (NDC), chemical formula) into the text entry field of the medical product search bar 502 and select the go button. When the go button is selected, the user computing device may send a medical product query or request including the medical product identifier to the provider computing system 104. In response, the provider computing system 104 may query or search the medical domain database 134 for information pertaining to the medical product identifier. If a result or multiple results are found, the provider computing system 104 may provide the results to the user computing device 112.

In this regard, the medical product search results section 504 is a section of the medical product group selection page 500 in which medical product search results received from the provider computing system 104 are displayed by the user computing device 112. Through interaction with the medical product search results section 504, the user of the user

computing device 112 can review and add one or more of the medical product search results to the medical product group. As shown, the medical product search results section 504 includes one or more medical product 506 search results (depending on the search results received from the provider computing system 104) and an add all button 515. The medical product search result 506 is shown to include a toggleable button 508, medical product information 512, and an add button 514.

The toggleable button 508 is a selectable button that hides or displays the medical product information of the respective medical product search result 506. In this regard, the user may select the toggleable button 508 to hide the medical product information 512 and to review the medical product search results 506 more quickly and then select the toggleable button 508 again to see the medical product information pertaining to the respective medical product search result 506 and do a more thorough review.

The medical product information 512 of the medical product search result 506 may be any medical product information described herein but is shown to include a medical product description or brand name, a medical product NDC, a chemical compound of the medical product, a company name, the strength of the medical product, the dose form of the medical product, and the annual count (number sold) of the medical product. As shown, each medical product search result 506 may comprise each type of the respective medical product (e.g., tablet version, solution version, different dosages, etc.) such that only one medical product search result 506 is returned for all forms of a single medical product.

Once the user has reviewed the medical product information 512 of the medical product search result 506 and determines they want to add the medical product to the medical product group, the user may select the add button 514. The add button 514 may add the medical product associated with the medical product search result 506 to the medical product group such that it appears in the medical product group summary 524. Similarly, the add all button 515 may add the respective medical product associated with each search medical product search result 506 of the medical product search results section 504 to the medical product group such that each medical product appears in the medical product group summary 524.

Still referring to FIG. 5, the filter options 516 are selectable options or checkboxes that, when selected, filter the medical product search results 506 of the medical product search results section 504. For example, the filter options 516 are shown to include a branded filter option 516, a generic filter option 516, a tablet filter option 516, and a solution filter option 516. If the user of the user computing device were to select the tablet filter option 516, the user computing device 112 may filter the medical product search results 506 and display only medical product search results 506 that are available as a tablet form (e.g., that include medical product information 512 indicating a tablet dose form) in the medical product search results section 504. Likewise, if the user were to select the branded filter option 516, the user computing device 112 may filter the medical product search results 506 and display only medical product search results 506 that are available as a branded medical product (e.g., that include medical product information 512 indicating a branded type of the medical product).

If at any time the user would like to review the medical products added to or currently included in the medical product group, the user can do so using the medical product group summary section 524. The medical product group

summary section **524** is a section of the medical product group selection page **500** that provides a summary of the medical products currently included in the medical product group. As shown, the medical product group summary section **524** includes multiple medical product titles or names **528** and a remove all button **532**. The medical product names **528** provide a quick idea of which medical products are in the medical product group and each include a remove button. The remove button (i.e., the “X” button on the righthand side of each medical product name **528**) is a selectable button that, when selected, removes the medical product identified by the medical product name **528** from the medical product group. Likewise, the remove all button **532** is a selectable button that, when selected, removes all of the medical products identified by the medical products names **528** from the medical product group.

Once the user has finished adding or removing medical products from the medical product group, the user can save the medical product group, cancel the changes to the medical product, or save the medical product group as a new group entirely. To do so, the user can interact with the save new medical group button **534**, the done/submit button **536**, or the cancel button **540**. The save new medical group button **534** is a selectable button that, when selected, generates a popup to name the medical group. Once the user has named the medical group, the user computing device **112** will provide the medical product group including the medical products added by the user to the provider computing system **104**. The provider computing system **104** may store the medical product group in association with an account of the user in the database **132** or the medical domain database **134**. The medical product group may then be used to filter the matched health and HCP information as described herein. Likewise, the done/submit button **536** is a selectable button **536** that, when selected, updates the currently access medical product group to include the medical products specified by the user and provides the updated medical product group including the medical products to the provider computing system **104** for storage as described herein. Lastly, the cancel button **540** is a selectable button that, when selected, discards any changes to the medical product group and returns the user to the HCP request page **400** (i.e., cancels the changes to the medical product group).

While the figures and description provided above are specific to the medical product group selection page **500**, the medical product groups, and medical products, it should be understood that a similar diagnoses group selection page (not shown) is accessible by selecting the diagnosing information filter **428** twice or by selecting the new filter button **430** under the diagnosing header. The diagnoses group selection page may include all of the same features and sections of the medical product group selection page **500** with application to the diagnoses groups described herein. For example, the user may access the diagnoses group selection page and be presented with a diagnoses search bar, a diagnosis search results section, a diagnoses group summary section, and so on. Through interaction with the diagnoses group selection page, the user may customize or generate new diagnoses groups that can be displayed and applied as filters via the diagnosing information filters **428**. Additionally, in some embodiments, the provider computing system **104** includes a diagnoses database that returns diagnoses search results in response to the user entering a diagnosis search in the diagnoses search bar.

Referring now to FIGS. 6A-6B, an HCP file generation request management page **600** (request management page **600**, for short) is shown. The request management page **600**

can be displayed on a display of the I/O circuit **180** of the user computing device **112** and, in general, provides the user with an interface (e.g., a page including options, information, and other features that will be described herein) to manage HCP file generation requests. For example, through interaction with the request management page **600** the user of the user computing device **112** can manage one or more HCP file generation requests including scheduling a time interval for HCP file generation requests and accessing, replicating, or deleting saved, but not submitted and generated, HCP file generation requests. To access or be navigated to the request management page **600**, the user of the user computing device **112** may select the job manager button **308**.

As shown, the request management page **600** includes the pull data button **304**, the job manager button **308**, the partner manager button **312**, the setup button **312**, a submitted request option or checkbox **604**, a draft (i.e., saved but not submitted) request option or checkbox **608**, and a request management section **610**. The submitted request option **604** and the draft request option **608** are each selectable options that manage the information displayed by the request management section **610**. For example, if the submitted request option **604** is selected, the request management section **610** may display request summaries **612** that represent HCP file generation request that have been generated and provided to the provider computing system **104**. In comparison, if the draft request option **608** is selected, the request management section **610** may display request summaries **612** of HCP file generation requests that have been saved (as discussed with regard to the saver buttons **404**) but not used for generation of an HCP file.

The request management section **610** may display HCP file generation request summaries **612** that each summarize an HCP file generation request that has been saved or used to generate an HCP file (depending on the selection of the option **604** or **608**). Each request summary **612** may include multiple request information fields such as a request ID field **616**, a request name field **620**, the request submission time **624**, a product field **628**, a request submitting account identifier field **632**, a request status field **640**, and a status time **644**. Each field of the request summary **612** will not be described in detail herein, but it should be understood that each field of the request summary **612** may be populated and received via the HCP request page **400** and may be included in an HCP file generation request.

To manage and filter the request summaries **612** that are shown and displayed in the request management section **610**, the request management page **600** is further shown to include multiple request summary filter options **668** or checkboxes and a request summary search bar **670**. The summary filter options **668** are selectable options or checkboxes that, when selected, filter the request summaries **612** displayed in the management section **610**. For example, the request management page **600** is shown to include a one time request summary filter option **668** and a scheduled request summary filter option **668**. If the user selects the scheduled summary filter option **668**, the user computing device may remove or not display request summaries **612** that only occurred one time/those that are not scheduled as will be described herein. Likewise, the request summary search bar **670** is a search bar through which the user can enter HCP file generation request identifiers (e.g., name, ID, date, etc.) and then select the search button. Once selected, only request summaries that match the identifier(s) will be displayed in the request management section **610**.

Each request summary 612 is further shown to include a duplicate request button 648, a schedule button 652 or edit schedule button 656, and a share button 660 or a cancel schedule button 664. The duplicate request button 648 is a selectable button that, when selected, may navigate the user to the HCP request page 400 having the same fields and options filled out as those of the HCP file generation request represented by the request summary 612. For example, if the HCP file generation request was generated using the HCP request page 400 having the "All Rheumatoid Arthritis" medical product information filter 432 selected, selecting the duplicate request button 648 may navigate the user to the HCP request page 400 having the "All Rheumatoid Arthritis" medical product information filter selected 432 as well. In this regard, each HCP file generation request represented by the request summary 612 may be substantially duplicated.

The schedule button 652 is a selectable button that, when selected, may generate a popup (not shown) to schedule the HCP file generation request represented by the request summary 612 to be generated or provided to the provider computing system 104 at a specific time interval or rate. For example, via the popup and after selecting the schedule button 652, the user of the user computing device 112 may indicate they want the HCP file generation request represented by the request summary 612 to be provided to the provider computing device 104 and a corresponding HCP file generated once a month, twice a month, once a week, every two days, etc. Likewise, if the HCP file generation request represented by the request summary 612 is already scheduled to be provided to the provider computing system 104 at a specific time interval or rate, the request summary 612 may include the edit schedule button 656. The edit schedule button 656 is a selectable button that, when selected, may provide the user with an interface (e.g., a popup) to edit the specific time interval or rate at which the HCP file generation request represented by the request summary 612 is provided to the provider computing system 104 (e.g., from every month to every two months). Additionally, the cancel schedule button 664 is a selectable button that may be selected to cancel the scheduling of the HCP file generation request represented by the request summary 612 and stop the providing of the HCP file generation request to the provider computing device 104 at a specific time interval or rate.

The share button 660 is a selectable button that, when selected, provides the user with an interface to share the generated HCP file with a recipient, as described herein. For example and referring to FIG. 6B, when the user of the user computing device 112 selects the share button 660, the HCP share popup 672 may be generated and displayed (e.g., overlaid) on the request management page 600. Through interaction with the HCP share popup 672, the user computing device 112 may generate a file share request and provide the file share request to the provider computing system 104. As shown, the HCP share popup 672 is shown to include a recipient drop down box 674, a recipient address text field 676, a license field 678, a project name field 680, a project description field 682, an expiration date/time field 684, a submit button 686, and a cancel button 688.

Each field of the HCP share popup 672 will not be described in detail but it should be understood that by entering a recipient address in the recipient address text field 676, the user decides and submits the recipient address of the file share request. Further, by entering an expiration date/time in the expiration date/time field 684, the user can decide the expiration date/time of the file share request.

Once the user has finished filling out each field of the HCP share popup 672, the user may select the submit button 686 or the cancel button 688. In response to the user selecting the submit button 686, the user computing device 112 may generate the file share request, including the recipient address of the recipient address text field 676 and the expiration date/time of the expiration date/time field 684, and provide the file share request to the provider computing system 104. In comparison, if the user selects the cancel button 688, the user computing device 112 may discard the information included within the HCP share popup 672 and close the HCP share popup 672.

Referring now to FIG. 7, an HCP file share management page 700 is shown. The file share management page 700 can be displayed on a display of the I/O circuit 180 of the user computing device 112 and, in general, provides the user with an interface (e.g., a page including options, information, and other features that will be described herein) to manage file share requests. For example, through interaction with the file share management page 700 the user of the user computing device 112 can manage one or more file share requests including altering (e.g., extending or shortening) the expiration date/time of the file share request. To access or be navigated to the file share management page 700, the user of the user computing device 112 may select the partner manager button 312.

As shown, the file share management page 700 includes a file share request management section 702, multiple file share request filter options 744, a reactivate button 748, and an end button 752. The file share request management section 702 may display multiple file share request summaries 704 that each summarize a file share request that has been provided to the provider computing system 104, and therefore used to provide the HCP file to the recipient address (e.g., the recipient computing device 116). Each file share request summary 704 may include multiple file share request fields such as the request ID field 616, the request name field 620, a project name field 708, a license field 712, a date/time of sharing field 716, a shared by account identifier field 720, a partner field 724, a shared with field 728, an expiration date/time field 732, and a status field 740. Each field of the file share request summary 704 will not be described in detail herein, but it should be understood that each field of the file share request summary 704 may be included in a file share request.

Each file share request summary 704 is further shown to include a schedule button 736 which is a selectable button that when selected, may generate a popup (not shown) to alter the expiration date/time of the file share request represented by the file share request summary 704. For example, via the popup or other interface generated when selecting the schedule button 736, the user of the user computing device 112 may indicate they want to extend the expiration date/time of the file share request by three days. In response, the user computing device 112 may indicate the new file share request expiration date/time to the provider computing system 104 or provide another file share request to the provider computing system 104. The recipient of the recipient computing device 116 associated with the recipient address may then access, download, and view the HCP file for an additional two days.

The file share request filter options 744 are selectable options or checkboxes that, when selected, filter the file share request summaries 704 shown within the file share request management section 704. For example, the file share management page 700 is shown to include an active file share request filter option 744 and an expired file share

41

request filter option **744**. If the user were to select the expired file share request filter option **744**, the user computing device may remove or not show file share request summaries **704** for file share requests that are active and show or display file share request summaries **704** for file share requests that are expired (e.g., as indicated by the status field **740**).

If a file share request is expired (e.g., the expiration date/time has passed) and the user wishes to reactivate it, the user may select the file share request summary **704** (e.g., by click on any of the fields of the file share request summary **704**) representing the file share request and then select the reactivate button **748**. The reactivate button **748** is a selectable button that, when selected, may generate a popup or other interface through which the user of the user computing device can reactivate the expired file share request. For example, the user may indicate a new expiration date/time for the expired file share request. In response, the user computing device **112** may generate a new file share request with the same information (e.g., recipient address) as the expired file share request and the new expiration date/time and provide it to the provider computing system **104**, which may then share the HCP file with the recipient computing device **116**. Likewise, if a file share request is active and the user wishes to end it, the user may select the file share request summary **704** representing the file share request and then select the end project button **752**. In response, the user computing device **112** may send an indication that the expiration date/time of the file share request is past and the HCP file is to not be shared with the recipient address.

Referring now FIGS. **8A-C**, an electronic file **800** (also referred to as an HCP file **800**) is shown according to an example embodiment. The electronic file **800** is shown to include NPI information for each HCP, a VID number for each HCP, HCP location information for each HCP (e.g., state, zip), specialty information for each HCP, HCP type for each HCP, number of diagnoses for each HCP, medical product prescription numbers for each medical product specified in a medical product group (e.g., drug **1**, drug **2**, drug **3**, and so on) and each HCP, new-to-brand prescriptions for each brand of medical product specified in the medical product group and each HCP, total number of patients seeing each HCP, payment type percentages for patients seeing each HCP (e.g., % of payments from Medicare, % of payments from Medicaid, % of payments from commercial health insurance, and % of payments in cash), and income range percentages of patients seeing each HCP.

As described herein, each electronic file may be a specified file type and be structured as selected by the user via the HCP request page **400**. For example, the electronic file **800** is shown to be a .XML file but may be other file types, as decided by the user through the file type options **476** of the HCP request page **400**. Further, while the electronic file **800** is shown to include new prescriptions and new-to-brand prescriptions HCP statistical values, the user may decide, via the medical product statistical value or insight options or checkboxes **456** of the HCP request page **400**, to include other HCP statistical values within the HCP file. In this regard, the electronic file **800** is just one example of many different possible structures and file types of the HCP file.

Referring now to FIGS. **9A-9B**, a method **900** for analyzing longitudinal health information and generating a dynamically structured electronic defined location prescribing file is shown, according to an example embodiment. While different overall, it should be understood that any steps or discussion of the method **900** may be applied or included within the method **200**, and vice versa, and that

42

such combinations are included within the scope of the present disclosure. For example, the method **200** may include any of the steps **904-956**, after or before any steps included in the method **200**, and the method **900** may include any of the steps **904-268**, after or before any of the steps included in the method **900**. Method **900** can be carried out by the system of FIG. **1**. More particularly, the method **900** can be carried out by the processing circuit **128** of the provider computing system **104** and through communication with the user computing device **112**, the covered entity computing system(s) **108**, and/or the recipient computing device **116**.

Method **900** commences at step **904** at which the provider computing system **104** receives or selects HCP information from the HCP database **124**. The HCP information may be associated with multiple HCPs and may include HCP location information for each of the HCPs. In some embodiments, at, before, or after step **904**, the provider computing system **104** may receive other information used to populate the health information request such as consumer information (from the consumer database **120**), timeframe information, medical product information (from the medical domain database **134**), and other information described herein as being included in the health information request. In some embodiments, the provider computing system **104** receives HCP information (or the other information with which the health information request is populated) from the HCP database **124** at specific time intervals (e.g., the same as or different than the specific intervals at which the health information requests are generated).

Once the provider computing system **104** has received the HCP information, the method **900** proceeds to step **908** at which the provider computing system **104** generates a health information request. As described herein, the health information request may be generated and populated to include the information received at step **904** such as the HCP information, the consumer information, the medical product information, and the timeframe information. The health information request **908** may request health information that is associated with or matches the information included in the health information request. In some embodiments, the health information request **908** may request health information that is not associated with or does not match the information included in the health information request.

In some embodiments, at step **908**, the provider computing system **104** may generate multiple health information requests. For example, the provider computing system **104** may generate a health information request for each covered entity computing system from which health information is sought. In other embodiments, the provider computing system **104** may generate a single, reused, health information request that is sent to each covered entity computing from which health information is sought.

Additionally, at step **908**, the provider computing system **104** may generate the health information request, at a specific time interval or rate (e.g., once a day, once a week, once a month, every 5 minutes, etc.). In some embodiments, the specific time interval or rate at which the health information request is generated is different than the specific time interval or rate at which the HCP information, medical product information, and/or consumer information is received. In other embodiments, the specific time interval or rate at which the health information request is generated is the same as the specific time interval or rate at which the HCP information, medical product information, and/or consumer information is received. By doing so, the provider computing system **104** may generate an updated health

information request including the most recently received HCP, medical product, and/or consumer information.

Once the provider computing system 104 has generated the health information request, the method 900 proceeds to step 912 at which the provider computing system 104 provides the health information request to one or more covered entity computing systems (e.g., the covered entity computing system 108). In some embodiments, the provider computing system 104 may provide the health information request over the network 118 via the network interface circuit 126 to the one or more covered entity computing systems 118.

Once the provider computing system 104 has provided the health information request to the one or more covered entity computing systems, the method 900 proceeds to step 916 at which health information is received from the one or more covered entity computing systems 108. The health information may be received in response to the provider computing system 104 providing the health information request to the one or more covered entity computing systems and may be or include Rx information, Mx information, medical product information, transaction information, diagnosis information, and other health information described herein. For example, the health information may be prescription claims (Rx) information that includes deidentified PHI, medical product information, diagnosis information (e.g., a diagnosis code), HCP information (e.g., an NPI of a prescribing HCP), date/time information, facility or health care system location information, and transaction information. In another example, the health information may be medical claims (Mx) information that includes deidentified PHI, medical product information, diagnosis information (e.g., a diagnosis code), HCP information (e.g., an NPI of a prescribing HCP), date/time information, facility or health care system location information, and transaction information. Further, the received health information may correspond to or be associated with the HCP information, the consumer information, and/or the medical product information included in the health information request. For example, the health information received at step 916 may be matched with or include consumer information for each individual consumer of each prescription. For example, each piece of Rx information may be matched with demographic information associated with the consumer of the prescription. Further, the received health information may include deidentified PHI, transaction information, date/time information, and medical product information.

In some embodiments, consumer information is received from the consumer database 120 at or after step 916 and matched with the HCP information. For example, the provider computing system 104 may receive consumer information from the consumer database 120 and match the consumer information with HCPs. In this regard, the consumer information may be organized by HCP such that HCP insights or statistical values described herein may be determined from the consumer information.

As described herein, the PHI may be deidentified, by the one or more covered entity computing systems 108, in a variety of ways including through encryption, tokenization, and using a hash function. In all cases, the one or more covered entity computing systems 108 may use a consistent form of deidentification such that the same PHI input will return the same hash value or token. For example, if a piece of health information included the name “John Smith,” a resulting deidentified hash value may be “1234321.” To be consistent, each of the one or more covered entity computing systems, when identifying the name “John Smith,” may then

return the hash value “123431.” By using the same hash function across the one or more covered entity computing systems, the provider computing system 104 can make additional connections between the received health information and generate HCP insights and statistical values. Further, less processing power is required by the provider computing system 104 as most of the health information matching is completed by the one or more covered entity computing systems prior to the health information being received by the provider computing system 104.

Once the provider computing system 104 has received the health information from the one or more covered entity computing systems 108, the method 900 proceeds to step 920 at which the provider computing systems 104 receives or selects government health information from the one or more national health databases 125. The government health information may include medical product information (e.g., NDC), location information (e.g., zip code for a specific Rx or Mx, state for a specific Rx or Mx, country for a specific Rx or Mx, etc.), deidentified PHI, and/or a timeframe (e.g., Jan. 1, 2022-Jan. 2, 2022) associated with the government health information. In some embodiments, the government health information includes Rx and/or Mx information and identifies whether a piece of information is Rx or Mx information. In other embodiments, the provider computing system 104 receives a first set of government health information that is Rx information and a second set of government health information that is Mx information.

As described herein, the provider computing system 104 may receive multiple sets of health information (e.g., at steps 920 and 916). For instance, the provider computing system 104 may receive first health information (e.g., at step 916) and second health information (e.g., at step 920). The first health information may be received from the one or more covered entity computing systems 108 and the second health information may be received from a health database (e.g., the national health database 125). In this regard, as described herein, the second health information may be or include government health information. In other embodiments, the order in which the health information is received may be switched. For instance, at step 916, the provider computing system 104 may receive first health information (which is or includes government health information) from the national health database 125, and at step 920, the provider computing system 104 may receive second health information from the one or more covered entity computing systems 108.

In some embodiments, the government health information is received from the one or more covered entity computing systems 108 at step 916. In some embodiments, the government health information does not include location information, and the provider computing system 104 matches or combines the government health information with the HCP location after step 916 (similar to or at step 928, as will be described further herein).

In some embodiments, after receiving the government health information and the health information, the provider computing system 104 remove duplicate health information and government health information, as discussed with regard to step 220 and 220 of the method 200.

Once the provider computing system 104 has received the government health information from the one or more covered entity computing systems 108, the method 900 proceeds to step 924 at which the provider computing system 104 selects or receives one or more digital projection models from the projection model database 135. In some embodiments, the provider computing system 104 may select one or more digital projection models based on the government

health information and the health information. For example, the provider computing system **104** may select a first digital projection model from the projection model database **135** based on the volume of the health information and/or the government health information (e.g., number of unique Rx claims, number of unique Mx claims, each piece of health information, etc.). Because different digital projection models have different use cases, the provider computing system **104** may select, at step **924**, the digital projection model that is best suited for the data set. For instance, prior to step **924**, the provider computing system **104** may determine the availability and quality of the received data/information (e.g., the combined health information, the government health information). Then, at step **924**, the provider computing system **104** may select specific models from the projection model database **135** based on the availability and quality of the received information. In one example, the provider computing system **104** may determine the received government health information is of poor quality (e.g., is outdated, is low volume, etc.) and select a projection model that only utilizes the combined health information. In another example, the provider computing system **104** may determine both sets of information are high quality and available. As a result, the provider computing system **104** may select a projection model that utilizes the combined health information and the government health information.

In some embodiments, at step **924**, the provider computing system **104** may select or receive each digital projection model included within the projection model database **135**. For instance, the provider computing system **104** may select a first digital projection model, a second digital projection model, and a third projection model. In some embodiments, the provider computing system **104** may select a first digital projection model from the projection model database **135** based on the health information, and the provider computing system **104** may select a second digital projection model from the digital projection model database **135** based on the government health information.

Once the provider computing system **104** has selected the one or more digital projection models from the projection model database **135**, the method **900** proceeds to step **928** at which the provider computing system **104** combines the HCP location information with the health information to generate combined health information. As described herein, the provider computing system **104** may match the HCP information with the health information by determining that one or more portions of the HCP information match that of the health information. For example, the provider computing system **104** may determine that the health information includes a national provider identifier (NPI) of a prescribing physician. The provider computing system **104** may then match the NPI of the prescribing physician with the NPI of the HCP information. In some embodiments, the NPI may be a hashed or tokenized NPI such that the health information may be sorted based on each NPI of each HCP. In other embodiments, the provider computing system **104** may match one or more of a HCP name, an HCP address, an HCP specialty, or an Rx number of the health information with that of the HCP information.

Once the provider computing system **104** has matched the HCP information with the health information, the provider computing system **104** may combine the location information of the HCP information with the matching health information to generate combined health information including the health information and location information. The combined health information may be sorted or categorized or sorted based on the location. For instance, the

combined health information may include a first portion (e.g., a column) grouped based on and identifiable with a first zip code, a second portion grouped based on and identifiable with a second zip code, a third portion grouped based on and identifiable with a first state, a fourth portion grouped based on and identifiable with a second state, a fifth portion grouped based on and identifiable with a first country, and/or a sixth portion grouped based on and identifiable with a second country.

In some embodiments, at or after step **928**, the provider computing system **104** combines the HCP location information with the received government health information to generate combined government health information. To combine the HCP location information with the received government health information, the provider computing system **104** may do the same as described with regard to the health information and step **928**.

In some embodiments, at step **928**, the provider computing system **104** may aggregate at least one of the health information, the government health information, the HCP information, the consumer information, and/or the location information. For instance, the provider computing system **104** may aggregate the health information, the government health information, the HCP information, the consumer information, and/or the location information based on specific aspects of the information. For example, the provider computing system **104** may aggregate the health information based on specific medical product, based on specific time-frame (e.g., the month of May 2021), based on specific location, and/or the like. For instance, the provider computing system **104** may receive a set of health information, from the one or more covered entity computing systems **108**. The health information, as described herein, may include a list of Rx or Mx claims information, deidentified PHI, and medical product information. Accordingly, the provider computing system **104**, at step **928**, may aggregate the health information into a count. For example, the provider computing system **104**, may aggregate all health information that is for the month of May 2020, Medical product X, the prescription type Rx, and the zip code 53202. The provider computing system **104** may then repeat this aggregation for each medical product, prescription type, and location.

In some embodiments, at or after step **928**, the provider computing system **104** may aggregate the government health information in the same way as the health information. In some embodiments, at or after step **928**, the provider computing system **104** may aggregate the government health information with the health information. In some embodiments, the provider computing system **104** may aggregate at least one of the health information, the government health information, the HCP information, the consumer information, and/or the location information prior to selecting the digital projection models at step **924**. In some embodiments, the provider computing system **104** may aggregate at least one of the health information, the government health information, the HCP information, the consumer information, and/or the location information based on the selected digital projection models. For instance, the provider computing system **104** may aggregate the at least one of the health information, the government health information, the HCP information, the consumer information, and/or the location information differently based on the selected digital projection model. For instance, a first ML digital projection model may request information aggregated differently than a second ML digital projection model.

Once the provider computing system **104** combined the HCP location information with the health information to

generate combined health information, the method **900** proceeds to step **936** at which the provider computing system **104** determines one or more sets of digitally projected values based on the combined health information and/or the government health information using the one or more digital projection models. As described herein, digital projection models are models (e.g., machine-learning (ML) models, statistical models, etc.) used to determine digitally projected prescription values (i.e., total projected prescription counts) based on the known health data (e.g., the combined health data, the government health data, etc.). In some embodiments, the digital projection models are ML models that utilize a feature data set to determine digitally projected prescription values. The feature data set may include the combined health information, the government health information, the HCP information, the consumer information, and/or other information described herein. Each set of digitally projected values may include multiple digitally projected values that are each associated with a different specific location. For instance, a set of digitally projected values may include a first digitally projected value that is associated with (i.e., projected specifically for) a first location (e.g., a first zip code, first state, first country); a second projected value that is associated with (i.e., projected specifically for) a second location (e.g., a second zip code, second state, second country), and so on. In this regard, each set of digitally projected values is associated with a specific location.

In some embodiments, the provider computing system **104** may determine a set of digitally projected values for multiple dates/times (e.g., day, month, year, week), for each medical product, and for each prescription type (e.g., Mx or Rx). For instance, the provider computing system **104** may determine a first set of digitally projected values for a first month, a first prescription type, and a first medical product; a second set of digitally projected values for a second month, a second prescription type and the first medical product; a third set of digitally projected values for a second month, a first prescription type, and a second medical product; and so on. In some embodiments, the provider computing system **104** may generate a first set of digitally projected values based on the government health information using a digital projection model and a second set of digitally projected values based on the government health information and the combined health information using the digital projection model. In other embodiments, the provider computing system **104** may generate a first set of digitally projected values based on the combined health information using the digital projection model. In another example, the government health information and the combined health information may each include a first portion associated with Rx claims and a second portion associated with Mx claims. Further, the provider computing system **104** may determine a first set of digitally projected values based on the first portion of the government health information and/or the first portion of the combined health information and a second set of digitally projected values based on the second portion of the government health information and/or the second portion of the combined health information.

In some embodiments, the provider computing system **104** may determine a digitally projected value using each digital projection model within the projection model database **135**. For instance, the digital projection model database **135** may include a first projection model (e.g., a binary classification projection/forecasting model), a second projection model (e.g., a CNN model), and a third projection model (e.g., a linear regression model). As a result, the

provider computing system **104** may determine a first digitally projected value using the first digital projection model, a second digitally projected value using the second digital projection model, and a third digitally projected value using the third digital projection model. Further, for each digitally projected value, the provider computing system **104** may determine a confidence score or value, a lower bound value, and/or a higher bound value. Then, the provider computing system **104** may rank each of the digitally projected values (e.g., based on the respective confidence score, based on a lower bound value, based on a higher bound value) and select the best digitally projected value (e.g., the highest confidence score that is above the lower bound value and below the higher bound value, the highest confidence score, etc.).

In this regard, the provider computing system **104** may select a final set of digitally projected values based on the rank of each digitally projected value. For instance, the provider computing system **104** may determine a first set of digitally projected values, a second set of digitally projected values, and a third set of digitally projected values, using a separate digital projection model for each set. Then, the provider computing system **104** may rank each respective (e.g., for the same projection (i.e., same medical product, same timeframe, same location(s), same prescription type)) digitally projected value of the first set of digitally projected values, a second set of digitally projected values, and a third set of digitally projected values. Then, based on the rankings, the provider computing system **104** may select a final set of digitally projected values, which could include any combination of the first, second, and third set of digitally projected values. For instance, the final set of digitally projected values, may include a first digitally projected value from the first set of digitally projected values, a second digitally projected values from the third set of digitally projected values, and so on.

In some embodiments, the lower bound value, and/or the higher bound value are received by the provider computing system (e.g., from the consumer database **120**, from the user computing device **112**, etc.). In some embodiments, the confidence score is determined based on the digital projection model, the government health data, and/or the combined health data.

Each digitally projected value may be a projected number of prescriptions for a specific date/time or timeframe, a specific medical product, a specific location, and/or a specific prescription type (e.g., Mx (in-office dispensed prescriptions claims) or Rx (pharmacy-dispensed prescription claims)). For example, because the one or more covered entity computing systems **112** do not cover or include health information associated with all possible prescriptions, the provider computing system **104** must project the missing portions of the health information. For instance, the covered entity computing systems **112** may include health information associated with 40% of all prescriptions in the United States. As a result, using the digital projection models, the provider computing system **104** may project the remaining 60% of prescriptions and determine the digitally projected values. Further, the provider computing system **104** may project the prescription numbers for each specific location (e.g., each zip code, each state, each country, etc.), for each medical product (e.g., Drug X, Vaccine Y, etc.), for each prescription type (e.g., number of Mx prescriptions and/or number of Rx prescriptions), and for each specific timeframe or date/time (e.g., for June 2021; for 2021; for Jul. 1, 2022, for the first week of June 2021, etc.). By doing so, the systems and methods described herein and the provider computing sys-

tem **104** provide a technical improvement to prescription projection systems. For instance, conventional prescription projection systems only consider Rx prescriptions and do not utilize or project Mx prescriptions. While this can be suitable for certain medical products, some medical products are primarily in-office dispensed (e.g., prescribed as Mx prescriptions (e.g., Intravenous (IV) fluids, Vaccines, etc.)). As a result, conventional prescription projection systems typically cannot accurately project prescription numbers for certain medical products. In comparison, the present systems and methods project the prescription numbers for each specific prescription type including Mx prescriptions. By doing so, the present systems and methods more accurately project total prescription numbers for medical products and provide more usable and accurate digitally projected values.

In some embodiments, at or after step **936**, the provider computing system **104** may further generate metadata associated with the set of digitally projected values. For instance, the provider computing system **104** may generate metadata indicating the set of digitally projected values is associated with a specific medical product, a specific timeframe, a specific location, a specific prescription type, and the like. In this regard, the metadata may be used to identify the set of digitally projected values. In other embodiments, the provider computing system **104** may generate metadata associated with each digitally projected value.

In some embodiments, at or after step **936**, the provider computing system **104** may store each set of digitally projected values (e.g., within the digitally projected value database (not shown)). For instance, the provider computing system **104** may add each set of digitally projected values to the digitally projected value database along with the metadata, described herein.

In some embodiments, the provider computing system **104** may generate a set of digitally projected values and use the set of digitally to determine another, summed, digitally projected value. For example, the provider computing system **104** may determine a set of digitally projected values that is associated with a specific medical product, a specific timeframe (e.g., the month of May 2021), a specification location (e.g., the zip code 53032), and a specific prescription type (e.g., Rx). In this regard, each digitally projected value may be associated with a specific day of the month of May 2021 (e.g., a first digitally projected value associated with May 1, 2021; a second digitally projected value associated with May 2, 2021; and so on). Then, to determine a summed digitally projected value for May 2021, the provider computing system **104** may sum the set of digitally projected values.

Once the provider computing system **104** has determined the one or more sets of digitally projected values, the method **900** proceeds to step **948** at which the provider computing system **104** receives a defined location prescribing file generation request from the user computing device **112**. As described herein, the defined location prescribing file generation request may indicate one or more insights or statistical values (e.g., median quantity dispensed of a medical product, total number of Rx prescriptions, total number of Mx prescriptions, the number of switches which occurred in the state, percentage of generic medical products prescribed or administered versus name brand medical products, etc.) as well as include a file destination address to which the HCP file is to be provided. Further, the defined location prescribing file generation request may define or identify one or more locations for which prescribing information is sought. In this regard, each defined location may include or be associated with multiple specific locations. For instance,

the defined location prescribing file generation request may identify one or more zip codes, one or more states, one or more countries, etc. In some embodiments, the defined location prescribing file generation request may also identify or include a specified file format that the defined location prescribing file is to be. For example, the defined location prescribing file generation request may indicate the defined location prescribing file is to be generated as a .CSV file delimited with commas.

In some embodiments, the defined location prescribing file generation request may also identify the medical product (s), a frequency, and/or a timeframe for which health information is sought. For example, the defined location prescribing file generation request may indicate a specific medical product (e.g., drug X), a specific timeframe (e.g., Jan. 1, 2021-Jan. 1, 2022), a type of prescriptions (e.g., Mx prescriptions and/or Rx prescriptions) that are sought, and a frequency (e.g., weekly) for which digitally projected values are sought and to be filtered.

Once the provider computing system **104** received the defined location prescribing file generation request from the user computing device **112**, the method **900** proceeds to step **950** at which the provider computing system **104** filters the digitally projected values based on the defined location prescribing file generation request. For example, the provider computing system **104** may filter the sets of digitally projected values based on the defined location prescribing file generation request by removing digitally projected values that do not match or are not associated with the medical product(s), defined location, frequency, and/or timeframe of the defined location prescribing file generation request. As described herein, each digitally projected value may be associated with a specific location (e.g., zip code 123, state X, country X, etc.), a specific prescription type (e.g., Mx or Rx), a specific timeframe or date/time (e.g., the month of January 2021; the year 2022; Jan. 1, 2021; etc.), and/or a specific medical product (e.g., drug X, Vaccine Y, etc.). In this regard, to filter the sets of digitally projected values, the provider computing system **104** may parse each set of digitally projected values and remove digitally projected values that do not match the specific location, specific timeframe, specific prescription type, and/or the specific medical product of the defined location prescribing file generation request. In some embodiments, to filter the sets of digitally projected values, the provider computing system **104** may parse each set of digitally projected values and select digitally projected values that match the specific location, specific timeframe, specific prescription type, and/or the specific medical product of the defined location prescribing file generation request.

As described herein, each set of digitally projected values may be associated with a specific medical product, timeframe, location, prescription type, etc. As a result, to filter the sets of digitally projected values, the provider computing system **104** may remove the sets of digitally projected values that do not match the specific location, specific timeframe, specific prescription type, and/or the specific medical product of the defined location prescribing file generation request. In some embodiments, to filter the sets of digitally projected values, the provider computing system **104** may parse each set of digitally projected values and select the sets of digitally projected values that do match the specific location, specific timeframe, specific prescription type, and/or the specific medical product of the defined location prescribing file generation request.

In some embodiments, at step **950**, the provider computing system **104** may select digitally projected values from

51

the digitally projected value database (not shown). Accordingly, the provider computing system **104** may filter the digitally projected values by only querying the digitally projected value database for a select set of the digitally projected values. For instance, at step **950**, the provider computing system **104** may select or query the digitally projected value database for digitally projected values that are associated with a specific timeframe, a specific medical product, a specific prescription type, a specific location, and the like. In this regard, the digitally projected value database may return values that match the query based on the meta-data of the digitally projected values meeting or being the same as the defined location prescribing file generation request (e.g., the same location, the same prescription type, etc.).

Once the provider computing system **104** has filtered the digitally projected values, the method **900** proceeds to step **952** at which the provider computing system **104** generates a defined location prescribing file including (e.g., populated with) at least one of the one or more sets of digitally projected values and/or the location information (e.g., of the combined health information and the government health information). For example, the provider computing system **104** may populate the defined location prescribing file with the filtered sets of digitally projected values or the filtered digitally projected values. Further, the provider computing system **104** may populate the defined location prescribing file with the locations that correspond or are associated with the filtered sets of digitally projected values or the filtered digitally projected values, the timeframes or date/times that correspond or are associated with the filtered sets of digitally projected values or the filtered digitally projected values, the specific prescription type(s) that are associated with the filtered sets of digitally projected values or the filtered digitally projected values, and/or the specific medical product that are associated with the filtered sets of digitally projected values or the filtered digitally projected values. FIG. **11** shows an example defined location prescribing file **1100**, and the respective fields populated by the provider computing system **104**.

Additionally, the provider computing system **104** may structure the defined location prescribing file based on the destination address of the defined location prescribing file generation request and/or the specific file type of the defined location prescribing file generation request. For example, the provider computing system **104** may populate the defined location prescribing file with the filtered digitally projected values and the location information in a specific order based on the defined location prescribing file generation request and the recipient address of the defined location prescribing file generation request. For instance, if the defined location prescribing file is to be provided to the user computing device **112** (e.g., the destination address is the IP address of the user computing device **112**), the provider computing system **104** may structure the defined location prescribing file to include a specific header for each set of digitally projected values and for each defined location. In comparison, if the defined location prescribing file is to be provided to another portion of the provider computing system **104** (e.g., Veeva Nitro), the provider computing system **104** may structure the HCP file to include a specific row for each set of digitally projected values and multiple rows for each location.

Moreover, the defined location prescribing file may be generated as a specific file type as indicated by the user computing device **112** in the defined location prescribing file. For example, the defined location prescribing file may

52

be generated as a CSV file, an Excel File (XLSX file), an XML file, a JSON file, and the like. By allowing the user (e.g., via the user computing device **112**) to decide the specific file type of the HCP file in the HCP file generation request and not only generating the defined location prescribing file as one specific file type, the provider computing system **104** may provide for improved shareability of the defined location prescribing file as well as use less processing power and memory. For example, in situations where the defined location prescribing file is only generated as a specific, non-dynamic, file type, third-party computing systems and recipient computing devices (e.g., the recipient computing device **116**) may have to perform additional file modification and conversion or request the provider computing system **104** to do so, which requires additional processing power, memory (to keep two files), and time. In comparison, the system **100** and the provider computing system **104** allow the user to specify the file type of the defined location prescribing file through the defined location prescribing file generation request. Accordingly, the user can specify the wanted file type the first time and the provider computing system **104** will generate the defined location prescribing file as the specified file type. As a result, no additional file modification or conversion is required reducing processing power usage, memory usage, and overall processing time for the provider computing system **104**.

Once the provider computing system **104** has generated the defined location prescribing file, the method **900** proceeds to step **956** at which the at which the HCP file is provided to the file destination address of the defined location prescribing file generation request. The file destination address may be any type of address or destination (e.g., IP address, web address, email address, file transfer protocol (FTP) address, and the like) to which the defined location prescribing file may be electronically transmitted or provided. In one example, the file destination address may be another server or circuit of the provider computing system **104** that is configured to receive the defined location prescribing file (e.g., Veeva Nitro®). In another example, the file destination address may be a web address of the Amazon Simple Storage Service® cloud storage web service. In another example, the file destination address may be the IP address of the user computing device **112** and the defined location prescribing file may be provided to the user computing device **112**.

Referring now to FIGS. **10A-10B**, user interfaces shown to the user of the user computing device **112** during the method **200** and/or the method **900** are shown, according to example embodiments. As described herein, the user interfaces of FIGS. **10-10B** may be one or more of web interfaces generated by the provider computing system **104** and rendered by the user computing device **112** as part of a web application or graphical user interfaces downloaded and generated by the user computing device **112** as part of a software application (e.g., mobile application, etc.). Further, the user interfaces shown on FIGS. **10A-10B** allow for communication between the user and the provider computing system **104** via the user computing device **112** (specifically via the I/O circuit **180**). Through interaction with the various user interfaces, the user may provide user input, feedback, and other information requested by the provider computing system **104**. In this regard, it should be understood that each interaction described herein by the user with the user interfaces of FIGS. **10A-10B** may be provided to the user computing device **112** and then transmitted to the provider computing system **104** and that each action described herein as occurring to or on the user computing

device **112** (e.g., navigating to a certain page, generating a popup, etc.) may be performed by the provider computing system **104**.

Still referring to FIGS. **10A-10B**, a defined location prescribing file generation request page **1000** (or prescribing file request page **1000**, for short), which can be displayed on a display of the I/O circuit **148** of the user computing device **112**, is shown. In general, the prescribing file request page **1000** provides the user with an interface to generate and provide a defined location prescribing file generation request to the provider computing system **104**. For example, through interaction with the prescribing file request page **1000** the user of the user computing device **112** can generate a defined location prescribing file indicating how the combined health information, the government health information, the sets of digitally projected values, the location information, and/or the consumer information should be filtered; identifying the file destination, identifying the file type of the defined location prescribing file; and indicating one or more statistical values or insights that are to be generated. To access or be navigated to the prescribing file request page **1000**, the user of the user computing device **112** may navigate to the information selection page **300** (e.g., by selecting the pull data button **304**), then select the prescription sales information option **328** or the national prescription information option **332**, and then select the next button **336**.

As shown, the prescribing file request page **1000** includes multiple save buttons **1004** (e.g., “Save” and “Save As”), a submit button **1008**, a medical product selection section **1012** (which may be similar to the filter section **424** of the HCP request page **400**), a location selection section **1022**, a prescription type selection section **1036**, a timeframe section **1048**, and a delivery menu **1072** (which may be similar to the delivery menu **472** of the HCP request page **400**).

The save buttons **1004** allow the user of the user computing device to save the current defined location prescribing file generation request such that they can return to it at another point in time. The “Save As” button may provide a popup menu or box through which the user can name the current HCP file generation request or overwrite another saved defined location prescribing file generation request. The “Save” button may simply save a pre-named or selected defined location prescribing file generation request such that the user can return to it under that pre-selected name at another point in time. When the user selects either of the save buttons **1004**, the user computing device **112** may generate an defined location prescribing file generation request, with the currently selected information and details of the prescribing file request page **1000** (including empty fields where the user has not made a selection yet) and provide the defined location prescribing file generation request to the provider computing system **104** with an indication that it is to be saved (but not used to generate an defined location prescribing file). The provider computing system **104** may receive the indication and the defined location prescribing file generation request and save the defined location prescribing file generation request (and accompanying selected information and name) in association with the account of the user. For example, the provider computing device may provide the defined location prescribing file generation request and accompanying information to the account database **132** for storage.

The medical product selection section **1012** is a section of the prescribing file request page **1000** through which the user of the user computing device **112** can customize and select the specific medical products to include in the defined location prescribing file generation request and to be used in

generating the defined location prescribing file. As shown, the medical products selection section **1012** includes an add products button **1014**, a select a template button **1016**, and multiple added product fields **1018**. The medical product selection section may be similar or the same as the medical product information filters **432** of the HCP file request page **400**. Accordingly, the add products button **1014**, the select a template button **1016**, and the multiple added product fields **1018** may be used to add medical products to the defined location prescribing file generation request, as described herein.

For instance, the user, via the prescribing file request page **1000**, may select one or more product fields or defined markets **1018** (e.g., Drug A, Drug B, All Rheumatoid Arthritis products, all Colon Cancer products, other medical product groups, etc.). Then, the user may select the add products button **1014** to add the products to the defined location prescribing file generation request, which may be provided to the provider computing system **104** (e.g., with the selection of the submit button **1008**) and used to filter the digitally projected values. In some embodiments, if the user selects the select a template button **1016**, the user may be navigated to the medical product group selection page **500**, where the user can generate and select one or more medical product groups.

The location selection section **1022** is a section of the prescribing file request page **1000** through which the user of the user computing device **112** can customize and select the specific locations to include in the defined location prescribing file generation request and to be used in generating the defined location prescribing file. As shown, the location selection section **1022** includes multiple location checkboxes or options **1024** and an add button **1032**. Accordingly, the multiple location checkboxes **1024** and the add button **1032** may be used to define (i.e., add locations to) the defined location of the defined location prescribing file generation request, as described herein.

For instance, the user, via the prescribing file request page **1000**, may select one or more location checkboxes **1024** (e.g., the state of Alabama, the state of Wisconsin, the zip code of 53039, the zip code of 53209, the zip code 53703, the country USA, the country Canada, etc.). Then, the user may select the add button **1032** to define the defined location and add the defined location to the defined location prescribing file generation request, which may be provided to the provider computing system **104** (e.g., with the selection of the submit button **1008**) and used to filter the digitally projected values.

The prescription type selection section **1036** is a section of the prescribing file request page **1000** through which the user of the user computing device **112** can customize and select the specific prescription types (e.g., Mx or Rx) to include in the defined location prescribing file generation request and to be used in generating the defined location prescribing file. As shown, the prescription type selection section **1036** includes a total prescriptions (TRx) checkbox **1040** and a total medical claims (TMx) checkbox **1042**. Accordingly, the TRx checkbox **1040** and the TMx checkbox **1042** may be used to add the specific prescription type(s) to the defined location prescribing file generation request, as described herein.

For instance, the user, via the prescribing file request page **1000**, may select the TRx checkbox **1040** and/or the TMx checkbox **1042**. The selected prescription type(s) (e.g., Mx and/or Rx) may then be added to the defined location prescribing file generation request and provided to the provider computing system **104** (e.g., with the selection of

the submit button **1008**). The selected prescription type(s) may then be used to filter the digitally projected values.

The timeframe selection section **1048** is a section of the prescribing file request page **1000** through which the user of the user computing device **112** can customize and select the specific timeframe to include in the defined location prescribing file generation request and to be used in generating the defined location prescribing file. As shown, the timeframe selection section **1048** includes multiple timeframe type checkboxes **1050**, a start date field **1052**, and an end date field **1054**. In some embodiments, the timeframe selection section **1048** may further include one or more frequency checkboxes (not shown). The frequency checkboxes may be selectable to add a specific requested frequency (e.g., daily, weekly, monthly, yearly) to the defined location prescribing file generation request, as described herein. Further, the timeframe type checkboxes **1050**, the start date field **1052**, the end date field **1054**, and/or the frequency checkboxes may be used to add the timeframe and/or the specified frequency to the defined location prescribing file generation request, as described herein.

For instance, the user, via the prescribing file request page **1000**, may select or set a start date via the start date field **1052** (e.g., Jun. 1, 2021) and select or set an end date via the end date field **1054** (e.g., Jun. 1, 2022), thereby setting a timeframe (e.g., Jun. 1, 2021-Jun. 1, 2022). The timeframe may then be added to the defined location prescribing file generation request and provided to the provider computing system **104** (e.g., with the selection of the submit button **1008**). The timeframe may then be used to filter the digitally projected values.

The delivery menu **1072** is a section of the prescribing file request page **1000** through which the user of the user computing device **112** can customize and select the file type and the file destination address to include in the defined location prescribing file generation request and to be used in generating the defined location prescribing file. As shown, the delivery menu **1072** includes a first checkbox **1076** and a second checkbox **1080**. The delivery menu **1072** may be similar or the same as the delivery menu **472** of the HCP file request page **400**. Accordingly, the first checkbox **1076** and the second checkbox **1080** may be used to add file type(s) and/or destination address(es) to the defined location prescribing file generation request, as described herein.

Referring now FIG. **11**, an electronic file **1100** (also referred to as a defined location prescriber file **1100**) is shown according to an example embodiment. The electronic file **1100** is shown to include a product name for each medical product and location, a month time frequency, a timeframe of Jan. 1, 2021 to Feb. 1, 2021, multiple locations (e.g., the zip code 90640, the zip code 00680, the zip code 08822, etc.), a prescription type of Rx, and a digitally projected value for each location.

The embodiments described herein have been described with reference to the drawings. The drawings illustrate certain details of specific embodiments that implement the systems, methods, and programs described herein. However, describing the embodiments with drawings should not be construed as imposing on the disclosure any limitations that may be present in the drawings.

It should be understood that no claim element herein is to be construed under the provision of 35 U.S.C. § 112(f), unless the element is expressly recited using the phrase “means for.”

As used herein, the term “circuit” may include hardware structured to execute the functions described herein. In some embodiments, each respective “circuit” may include

machine-readable media for configuring the hardware to execute the functions described herein. The circuit may be embodied as one or more circuitry components including, but not limited to, processing circuitry, network interfaces, peripheral devices, input devices, output devices, sensors, etc. In some embodiments, a circuit may take the form of one or more analog circuits, electronic circuits (e.g., integrated circuits (IC), discrete circuits, system on a chip (SOC) circuits), telecommunication circuits, hybrid circuits, and any other type of “circuit.” In this regard, the “circuit” may include any type of component for accomplishing or facilitating achievement of the operations described herein. For example, a circuit as described herein may include one or more transistors, logic gates (e.g., NAND, AND, NOR, OR, XOR, NOT, XNOR), resistors, multiplexors, registers, capacitors, inductors, diodes, wiring, and so on.

The “circuit” may also include one or more processors communicably coupled to one or more memory or memory devices. In this regard, the one or more processors may execute instructions stored in the memory or may execute instructions otherwise accessible to the one or more processors. In some embodiments, the one or more processors may be embodied in various ways. The one or more processors may be constructed in a manner sufficient to perform at least the operations described herein. In some embodiments, the one or more processors may be shared by multiple circuits (e.g., circuit A and circuit B may comprise or otherwise share the same processor which, in some example embodiments, may execute instructions stored, or otherwise accessed, via different areas of memory). Alternatively or additionally, the one or more processors may be structured to perform or otherwise execute certain operations independent of one or more co-processors. In other embodiments, two or more processors may be coupled via a bus to enable independent, parallel, pipelined, or multi-threaded instruction execution. Each processor may be implemented as one or more general purpose processors, application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), digital signal processors (DSPs), or other suitable electronic data processing components structured to execute instructions provided by the memory. The one or more processors may take the form of a single core processor, a multi-core processor (e.g., dual core, quad core, etc.), micro-processor, etc. In some embodiments, the one or more processors may be external to the apparatus. For example, the one or more processors may be a remote processor (e.g., a cloud-based processor). Alternatively or additionally, the one or more processors may be internal and/or local to the apparatus. In this regard, a given circuit or components thereof may be disposed locally (e.g., as part of a local server, a local computing system) or remotely (e.g., as part of a remote server such as a cloud-based server). To that end, a “circuit” as described herein may include components that are distributed across one or more locations.

An example system for implementing the overall system or portions of the embodiments might include a general purpose computing device in the form of computers, including a processing unit, a system memory, and a system bus that couples various system components including the system memory to the processing unit. Each memory device may include non-transient volatile storage media, non-volatile storage media, non-transitory storage media (e.g., one or more volatile and/or non-volatile memories), etc. In some embodiments, the non-volatile storage media may take the form of ROM, flash memory (e.g., flash memory such as NAND, 3D NAND, NOR, 3D NOR), EEPROM, MRAM, magnetic storage, hard disks, optical disks, etc. Combina-

tions of the above are also included within the scope of machine-readable media. In this regard, machine-executable instructions comprise, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machine to perform a certain function or group of functions. Each respective memory device may be operable to maintain or otherwise store information relating to the operations performed by one or more associated circuits, including processor instructions and related data (e.g., database components, object code components, script components), in accordance with the example embodiments described herein.

It should also be noted that the term “input devices,” as described herein, may include any type of input device including, but not limited to, a keyboard, a keypad, a mouse, a joystick, or other input devices performing a similar function. Comparatively, the term “output device,” as described herein, may include any type of output device including, but not limited to, a computer monitor, printer, facsimile machine, or other output devices performing a similar function.

It should be noted that although the diagrams herein may show a specific order and composition of method steps, it is understood that the order of these steps may differ from what is depicted. For example, two or more steps may be performed concurrently or with partial concurrence. Also, some method steps that are performed as discrete steps may be combined, steps being performed as a combined step may be separated into discrete steps, the sequence of certain processes may be reversed or otherwise varied. The order or sequence of any element or apparatus may be varied or substituted according to alternative embodiments. Accordingly, all such modifications are intended to be included within the scope of the present disclosure as defined in the appended claims. Such variations will depend on the machine-readable media and hardware systems chosen and on designer choice. It is understood that all such variations are within the scope of the disclosure. Likewise, software and web implementations of the present disclosure could be accomplished with standard programming techniques with rule-based logic and other logic to accomplish the various database searching steps, correlation steps, comparison steps, and decision steps.

The foregoing description of embodiments has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure to the precise form disclosed, and modifications and variations are possible in light of the above teachings or may be acquired from this disclosure. The embodiments were chosen and described in order to explain the principles of the disclosure and its practical application to enable one skilled in the art to utilize the various embodiments and with various modifications as are suited to the particular use contemplated. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions and embodiment of the embodiments without departing from the scope of the present disclosure as expressed in the appended claim.

What is claimed is:

1. A method for generating a dynamically structured electronic file in a dynamic file generation system, wherein the dynamic file generation system comprises a provider computing system, one or more covered entity computing systems, and a user computing device associated with a user connected by a secure network, the method comprising:

receiving, by a network interface circuit of the provider computing system, first health information including location information from a national health database; generating, by a processing circuit of the provider computing system, a health information request; providing, by the network interface circuit of the provider computing system, the health information request to the one or more covered entity computing systems; receiving, by the network interface circuit, second health information including prescription claim (Rx) health information and medical claim (Mx) health information from the one or more covered entity computing systems, wherein the Rx health information includes first deidentified protected health information (PHI), and wherein the Mx health information includes second deidentified protected health information (PHI); receiving, by the network interface circuit, health care provider (HCP) information including HCP location information; combining, by the processing circuit, the Rx health information with at least a portion of the HCP location information to generate first combined health information; combining, by the processing circuit, the Mx health information with at least a portion of the HCP location information to generate second combined health information; selecting, by the processing circuit, one or more digital projection models from a projection model database of the provider computing system; determining, by the processing circuit, a first plurality of digitally projected values based on at least one of the first combined health information or the Rx health information using at least one of the one or more digital projection models, wherein each digitally projected value of the first plurality of digitally projected values is projected for prescription claims and associated with time information; determining, by the processing circuit, a second plurality of digitally projected values based on at least one of the second combined health information or the Mx health information using at least one of the one or more digital projection models, wherein each digitally projected value of the second plurality of digitally projected values is projected for medical claims and associated with time information; receiving, by the network interface circuit, a file generation request from the user computing device, the file generation request identifying a file type, a timeframe, and a file destination address; filtering, by the processing circuit, the first plurality of digitally projected values for the timeframe of the file generation request by removing digitally projected values with time information that are not within the timeframe; filtering, by the processing circuit, the second plurality of digitally projected values for the timeframe of the file generation request by removing digitally projected values with time information that are not within the timeframe; generating, by the processing circuit, the electronic file including at least one digitally projected value of the filtered first plurality of the digitally projected values, at least one digitally projected value of the filtered second plurality of the digitally projected

59

values, and at least a portion of the HCP location information or the location information, wherein the electronic file is generated as the file type identified by the file generation request, and wherein the electronic file is structured based on at least one of the file generation request or the file destination; and

providing, by the network interface circuit, the electronic address to the file destination address of the file generation request.

2. The method of claim 1, further comprising: receiving, by the network interface circuit, a file share request from the user computing device, the file share request including a recipient address and a file share expiration date;

receiving, by the network interface circuit, a request to access the electronic file from a recipient computing device associated with the recipient address;

determining, by the processing circuit, that a present date is before the expiration date;

providing, by the network interface circuit and in response to determining that the present date is before the expiration date, the electronic file to the recipient computing device associated with the recipient address based on the request to access the electronic file.

3. The method of claim 2, further comprising: generating, by the processing circuit and in response to receiving the file share request, an indication comprising a uniform resource locator (URL) including an expiration parameter that deactivates the URL on the file share expiration date; and

providing, by the network interface circuit, the indication to the recipient computing device associated with the recipient address.

4. The method of claim 1, wherein each digital projection model of the one or more digital projections models is at least one of: a binary classification model, a neural network model, a multiclass classification model, or a regression model.

5. The method of claim 1, wherein the one or more digital projection models is a plurality of digital projection models, wherein the first plurality of digitally projected values is a plurality of sets of digitally projected values, and wherein the provider computing system determines a confidence score for each digitally projected value of a set of digitally projected values using each digital projection model.

6. The method of claim 5, further comprising: ranking, by the processing circuit, each digitally projected value based on the confidence score;

selecting, by the processing circuit, a final set of digitally projected values based on the rankings; and

filtering, by the processing circuit, the final set of digitally projected values for the defined location of the file generation request by removing digitally projected values that are not associated with the defined location.

7. The method of claim 1, wherein the one or more covered entity computing systems comprise a first covered entity computing system associated with a pharmacy and a second covered entity computing system associated with a medical claims company, wherein the combined health information is first combined health information, and wherein the method further comprises:

receiving, by the network interface circuit, the Rx health information from the first covered entity computing system;

60

receiving, by the network interface circuit, the Mx health information from the second covered entity computing system.

8. The method of claim 7, further comprising: determining, by the processing circuit, the first combined health information includes duplicate health information; and

modifying, by the processing circuit, the first combined health information to remove the duplicate health information.

9. The method of claim 1, wherein each digitally projected value of the plurality of digitally projected values is associated with a specific location, wherein the file generation request identifies a defined location, and wherein the method further comprises:

filtering, by the processing circuit, the first plurality of digitally projected values for the defined location of the file generation request by removing digitally projected values that are not associated with the defined location.

10. The method of claim 9, wherein the defined location includes a plurality of locations, and wherein each location of the plurality of locations is at least one of: a country, a state, or a zip code.

11. The method of claim 1, wherein the first health information is national health insurance information associated with at least one of Medicaid or Medicare.

12. A method for generating a dynamically structured electronic file in a dynamic file generation system, wherein the dynamic file generation system comprises a provider computing system, one or more covered entity computing systems, and a user computing device connected by a secure network, the method comprising:

receiving, by a network interface circuit of the provider computing system, first health information including location information from a national health database;

receiving, by the network interface circuit, second health information including prescription claim (Rx) health information and medical claim (Mx) health information from the one or more covered entity computing systems,

wherein the Rx health information includes first deidentified protected health information (PHI), and

wherein the Mx health information includes second deidentified protected health information (PHI);

receiving, by the network interface circuit, health care provider (HCP) information including HCP location information;

combining, by the processing circuit, the Rx health information with at least a portion of the HCP location information to generate first combined health information;

combining, by the processing circuit, the Mx health information with at least a portion of the HCP location information to generate second combined health information;

selecting, by the processing circuit, one or more digital projection models from a projection model database of the provider computing system;

determining, by the processing circuit, a first set of digitally projected values based on the first health information using the one or more digital projection models;

determining, by the processing circuit, a second set of digitally projected values based on the first combined health information and the first health information using at least one of the one or more digital projection models,

61

wherein each digitally projected value of the first set of digitally projected values associated with time information, wherein each digitally projected value of the second set of digitally projected values is projected for prescription claims and associated with time information; 5

determining, by the processing circuit, a third set of digitally projected values based on the second combined health information and the first health information using at least one of the one or more digital projection models, 10

wherein each digitally projected value of the third set of digitally projected values is projected for medical claims and associated with time information;

receiving, by the network interface circuit, a file generation request from the user computing device, the file generation request identifying a file type, a timeframe, and a file destination address; 15

filtering, by the processing circuit, the first set of digitally projected values for the timeframe of the file generation request by removing digitally projected values with time information that are not within the timeframe; 20

filtering, by the processing circuit, the second set of digitally projected values for the timeframe of the file generation request by removing digitally projected values with time information that are not within the timeframe; 25

filtering, by the processing circuit, the third set of digitally projected values for the timeframe of the file generation request by removing digitally projected values with time information that are not within the timeframe; 30

generating, by the processing circuit, the electronic file including at least one digitally projected value of the filtered first set of digitally projected values, the second set of digitally projected values, or the filtered second set of digitally projected values, wherein the electronic file includes at least a portion of the location information or the HCP location information, 35

wherein the electronic file is generated as the file type identified by the file generation request, and 40

wherein the electronic file is structured based on at least one of the file generation request or the file destination; and

providing, by the network interface circuit, the electronic address to the file destination address of the file generation request. 45

13. The method of claim 12, wherein each digitally projected value of the first set of digitally projected values and the second set of digitally projected values is associated with a specific location, wherein the file generation request identifies a defined location, and wherein the method further comprises: 50

filtering, by the processing circuit, the first set of digitally projected values and the second set of digitally projected values for the defined location of the file generation request by removing digitally projected values that are not associated with the defined location. 55

62

eration request by removing digitally projected values that are not associated with the defined location.

14. The method of claim 13, wherein the defined location includes a plurality of locations, and wherein each location of the plurality of locations is at least one of: a country, a state, or a zip code.

15. The method of claim 12, wherein the first health information is government health information comprising a first portion associated with prescription claims and a second portion associated with medical claims.

16. The method of claim 15, wherein the first set of digitally projected values comprises a first plurality of digitally projected values and a second plurality of digitally projected values, and wherein determining the first set of projected values comprises: 5

determining, by the processing circuit, the first plurality of digitally projected values based on the first portion of the government health information using at least one of the one or more digital projection models; and

determining, by the processing, the second plurality of digitally projected values based on the second portion of the government health information using at least one of the one or more digital projection models.

17. The method of claim 12, wherein each digital projection model of the one or more digital projections models is at least one of: a binary classification model, a neural network model, a multiclass classification model, or a regression model.

18. The method of claim 12, wherein the one or more covered entity computing systems comprise a first covered entity computing system associated with a pharmacy and a second covered entity computing system associated with a medical claims company, wherein the combined health information is first combined health information, and wherein the method further comprises: 10

receiving, by the network interface circuit, the Rx health information from the first covered entity computing system;

receiving, by the network interface circuit, the Mx health information from the second covered entity computing system.

19. The method of claim 12, wherein each digitally projected value of the first set of digitally projected values and each digitally projected value of the second set of digitally projected values is associated with a specific month and a specific year, wherein the first set of digitally projected values are filtered to remove digitally projected values in which the specific month and the specific year do not fall within the timeframe, and wherein the second set of digitally projected values are filtered to remove digitally projected values in which the specific month and the specific year do not fall within the timeframe. 15

* * * * *